Category Theory

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Contents

1 Equivalences **1**

2 Limits **5**

2.1 First examples of (co)limits . . . 5
2.2 In general . . . 5
2.3 More examples . . . 6
2.4 Limits as products and equalizers . . . 6
2.5 Group actions . . . 9
2.6 Pullbacks and pushforwards . . . 9
2.7 Preservation of (co)limits . . . 11
2.8 Limits in functor categories . . . 16
2.9 Functorality of limits and colimits . . . 17

3 Yoneda's Lemma **18**

3.1 Representable functors . . . 18
3.2 Yoneda's lemma . . . 18
3.3 Representables and (co)limits . . . 20

4 Adjunctions **22**

4.1 The unit and counit . . . 23
4.2 Equivalences and adjunctions . . . 26
4.3 Adjunctions and (co)limits . . . 27
4.4 (Co)limits as adjunctions . . . 28
4.5 Per-object adjunctions . . . 29
4.6 The adjoint functor theorem . . . 31

5 Abelian Categories **33**

5.1 Zero morphisms, zero objects . . . 33
5.2 Preadditive categories . . . 35
5.3 (Co)kernels, monomorphisms, epimorphisms . . . 36

6 Monads **38**

6.1 Monads from adjunctions . . . 38
6.2 Adjunctions from monads . . . 40
6.3 Monadic functors . . . 45
6.4 Canonical presentations via free algebras . . . 46
6.5 Recognizing categories of algebras . . . 48
6.6 Limits and colimits in categories of algebras . . . 50

I begin by noting that these notes are for those already familiar with the very basic definitions of category theory (those of categories, functors, and natural transformations).

1 Equivalences

1.0.1 Definition

Let $F, G: \mathcal{C} \to \mathcal{D}$ be functors and $\eta: F \Rightarrow G$ be a natural transformation.

(1) Let $H: \mathcal{D} \to \mathcal{E}$ be another functor, then $H\eta: HF \Rightarrow HG$ defined by $(H\eta)_c = H(\eta_c)$ is a natural transformation.

(2) Let $K: \mathcal{E} \to \mathcal{C}$ be another functor, then $\eta H: FH \to GH$ defined by $(\eta H)_e = \eta_{Hc}$ is a natural transformation.

These operations on natural transformations are called **whiskering**.

1.0.2 Definition

Let $\eta: F \Rightarrow G$ and $\epsilon: G \Rightarrow H$ be natural transformations between functors $\mathcal{C} \to \mathcal{D}$, then their **vertical composition** is a natural transformation $\epsilon \circ \eta: F \Rightarrow H$ defined to be $(\epsilon \circ \eta)_X = \epsilon_X \circ \eta_X$. This is called *vertical* composition as we can view F, G, H as being stacked vertically and η, ϵ as arrows pointing vertically between them, their vertical composition is taken by following these arrows.

Similarly if $\eta: F \Rightarrow G$ is natural between functor $\mathcal{C} \to \mathcal{D}$ and $\epsilon: J \Rightarrow K$ between functors $\mathcal{D} \to \mathcal{E}$, then their **horizontal composition** is a natural transformation $\epsilon * \eta: JF \Rightarrow KG$ defined by $(\epsilon * \eta)_X = \epsilon G \circ J\eta = K\eta \circ \epsilon F$.

Note that for natural transformations $\alpha, \alpha', \beta, \beta'$ we have the **interchange law**:

$$(\beta' \circ \alpha') * (\beta \circ \alpha) = (\beta' * \beta) \circ (\alpha' * \alpha)$$

1.0.3 Definition

An **equivalence** of two categories $\mathcal{C}, \mathcal{D}$ is a pair of functors $F: \mathcal{C} \rightleftarrows \mathcal{D}: G$ such that the compositions are naturally isomorphic to the identity functors on the respective categories. That is, $GF \cong 1_{\mathcal{C}}$ and $FG \cong 1_{\mathcal{D}}$. Two categories are **equivalent** if there exists an equivalence between them.

Clearly the equivalence of categories forms an equivalence relation; transitivity is due to the composition of equivalences being an equivalence.

1.0.4 Definition

A functor $F: \mathcal{C} \to \mathcal{D}$ is:

(1) **full** if for every $x, y \in \mathcal{C}$, the map $\mathcal{C}(x, y) \to \mathcal{D}(Fx, Fy)$, $f \mapsto Ff$ is surjective;

(2) **faithful** if for every $x, y \in \mathcal{C}$, the map $\mathcal{C}(x, y) \to \mathcal{D}(Fx, Fy)$ is injective;

(3) **essentially surjective** if for every $d \in \mathcal{D}$ there exists a $c \in \mathcal{C}$ such that Fc is isomorphic to d.

A full and faithful functor is called **fully faithful**.

1.0.5 Theorem

A functor in an equivalence is fully faithful and essentially surjective. Conversely any fully faithful and essentially surjective function forms an equivalence.

Proof

F being fully faithful means that $\mathcal{C}(x,y) \to \mathcal{D}(Fx,Fy)$ is a bijection. Indeed, if G is the other component in the equivalence then we have a composition of maps

$$\mathcal{C}(x,y) \xrightarrow{F} \mathcal{D}(Fx,Fy) \xrightarrow{G} \mathcal{C}(GFx,GFy) \cong \mathcal{C}(x,y)$$

where the last bijection is induced by the natural isomorphism $\epsilon: GF \cong 1_{\mathcal{C}}$. This maps $f: x \to y$ to $\epsilon_Y GF f \epsilon_X^{-1}$, which is just equal to f by naturality. Composition in the other direction gives the identity as well, so our map is bijective as desired. Essential surjectivity is clear: $FGx \cong x$ for all $x \in \mathcal{D}$.

For the converse we construct $G: \mathcal{D} \to \mathcal{C}$ as the other component in the equivalence. Let $d \in \mathcal{D}$, then by essential surjectivity there exists $x \in \mathcal{C}$ such that $Fx \cong d$. Let us choose such an x, and define $Gd = x$. And for $g: d \to d'$, as we have $\alpha: d \cong Fx, \alpha': d' \cong Fx'$ then $\alpha' g \alpha^{-1}: Fx \to Fx'$. By full faithfulness, there is a unique $f: x \to x'$ such that $Ff = g$, so define $Gg = f$. This can be seen to be a functor, and we can see that F, G forms an equivalence. ■

1.0.6 Proposition

Fully faithful functors reflect isomorphisms: if $F: \mathcal{C} \to \mathcal{D}$ is fully faithful and Ff is an isomorphism, then so is f.

Proof

Suppose $f: c \to d$ such that Ff is an isomorphism. Then there is a unique $g: d \to c$ such that Fg is its inverse, by full faithfulness. Then we have that $F(fg), F(gf)$ are both the identities, but since F is faithful this means that fg and gf must be the identities, so $f^{-1} = g$. ■

1.0.7 Definition

A category is **connected** if every finite full subcategory is connected as an undirected graph. That is to say, for every two objects X, Y there is a finite sequence of objects $X = X_0, \ldots, X_n = Y$ such that $\hom(X_i, X_{i+1})$ or $\hom(X_{i+1}, X_i)$ is non-empty.

1.0.8 Corollary

Any connected groupoid is equivalent to the automorphism group of any of its objects.

Proof

A connected groupoid is one for which there exists isomorphisms between any two objects. Let $g \in \mathcal{G}$ and let $G = \mathcal{G}(g,g)$ denote its automorphism group. Then the functor $\mathsf{B}G \to \mathcal{G}$ mapping the unique object of $\mathsf{B}G$ to g and the elements of G to their respective elements in $G = \mathcal{G}(g,g)$ is fully faithful and essentially surjective (as all objects in $\mathcal{G}$ are isomorphic). Thus this forms an equivalence. ■

1.0.9 Definition

Let $\mathcal{C}$ be a category, its **skeleton** is the subcategory $\mathsf{Sk}\mathcal{C} \subseteq \mathcal{C}$ obtained by choosing a single object from each isomorphism class in $\mathcal{C}$.

Note that we say *the* skeleton of $\mathcal{C}$ even though the construction of $\mathsf{Sk}\mathcal{C}$ depends on choosing a representative from each isomorphism class. But it is not hard to see that any choice gives isomorphic categories, thus $\mathsf{Sk}\mathcal{C}$ is defined uniquely up to isomorphism.

1.0.10 Proposition

The inclusion $\mathsf{Sk}\mathcal{C} \to \mathcal{C}$ is an equivalence of categories.

Proof

This is clear, since $\mathsf{Sk}\mathcal{C}$ is a full subcategory the inclusion is fully faithful. By definition of $\mathsf{Sk}\mathcal{C}$ the inclusion is also essentially surjective, as desired. ■

For a connected groupoid, we clearly have that $\mathsf{Sk}\mathcal{G}$ is simply a category with a single object whose hom-set is the automorphism group of any object in $\mathcal{G}$. That is, $\mathsf{Sk}\mathcal{G} = \mathsf{B}\mathcal{G}(g, g)$ for any $g \in \mathcal{G}$. So our previous claim is just a special example of the above proposition.

1.0.11 Example

Let $\mathcal{C}$ be a category equivalent to the trivial category $[0]$. Then $F\colon \mathcal{C} \to [0]$ is fully faithful, which means that $\hom(x, y) \cong \hom_0(0, 0)$, i.e. it is a singleton. So between every two objects in $\mathcal{C}$ there is a unique morphism, in particular it must be an isomorphism.

In general if $\mathcal{C}$ is equivalent to a groupoid, it is a groupoid. Notice that if $F\colon \mathcal{C} \to \mathcal{D}$ is an equivalence then every $\mathcal{D}$-morphism is of the form $\alpha \circ Ff \circ \beta$ for α, β isomorphisms and f a $\mathcal{C}$-morphism. So if $F\colon G \to \mathcal{C}$ is an equivalence then every $\mathcal{C}$-morphism is of the form $\alpha \circ Fg \circ \beta$, and since g is an isomorphism so is Fg, so every $\mathcal{C}$-morphism is an isomorphism.

1.0.12 Proposition

For categories $\mathcal{C}, \mathcal{D}$ the following are equivalent:

(1) $\mathcal{C}$ and $\mathcal{D}$ are equivalent,

(2) $\mathsf{Sk}\mathcal{C}$ and $\mathsf{Sk}\mathcal{D}$ are isomorphic,

(3) $\mathsf{Sk}\mathcal{C}$ and $\mathsf{Sk}\mathcal{D}$ are equivalent.

Proof

If $F\colon \mathsf{Sk}\mathcal{C} \to \mathsf{Sk}\mathcal{D}$ is an equivalence, then it must be surjective by essential surjectivity. Furthermore it must be injective: if $F(c) = F(c')$ then in particular $F(c), F(c')$ are isomorphic and since F reflects isomorphisms, $c \cong c'$. Thus $c = c'$ since the skeleton has a unique object of each isomorphism class. This means that F is an isomorphism, so (2) and (3) are equivalent.

If $F\colon \mathcal{C} \to \mathcal{D}$ is an equivalence then it restricts to an equivalence $\mathsf{Sk}\mathcal{C} \to \mathsf{Sk}\mathcal{D}$ (as it is still fully faithful and essentially surjective). And if $\mathsf{Sk}\mathcal{C}$ and $\mathsf{Sk}\mathcal{D}$ are equivalent, since $\mathcal{C}$ and $\mathcal{D}$ are equivalent to their skeletons, by transitivity $\mathcal{C}$ and $\mathcal{D}$ are equivalent. ■

Thus if $\mathcal{C}$ is equivalent to $[0]$, then $\mathsf{Sk}\mathcal{C} \cong \mathsf{Sk}[0] = [0]$, i.e. it has a single isomorphism class, as we already showed. And if $\mathcal{C}$ is equivalent to a groupoid $\mathcal{G}$, then $\mathsf{Sk}\mathcal{C} \cong \mathsf{Sk}\mathcal{G}$ which has a single object whose hom-set is only isomorphisms. Thus $\mathcal{C}$ must be a groupoid: it has a single isomorphism class and all of its morphisms are isomorphisms.

1.0.13 Example

Let k be a field, then we have a category $\mathrm{Vec}_k^{\mathsf{fin}}$ of finite-dimensional k-vector spaces. We also have Mat_k, the category of matrices over k defined as follows:

(1) Objects in Mat_k are natural numbers (including 0),

(2) Morphisms $n \to m$ are matrices of dimension $m \times n$, i.e. $\mathrm{Mat}_k(n, m) = \mathrm{Mat}_{m\times n}(k)$,

(3) For matrices $A\colon n \to m$ and $B\colon m \to p$, define their composition to be their multiplication: $B \circ A = BA$.

These categories are equivalent: we define a functor $F\colon \mathrm{Mat}_k \to \mathrm{Vec}_k^{\mathsf{fin}}$ by $F(n) = k^n$ and for $A\colon n \to m$, $F(A)\colon k^n \to k^m, v \mapsto Av$. Simple linear algebra shows that this is essentially surjective and fully faithful. Note that $\mathsf{Sk}\mathrm{Mat}_k = \mathrm{Mat}_k$, and thus by equivalence $\mathsf{Sk}\mathrm{Vec}_k^{\mathsf{fin}} \cong \mathrm{Mat}_k$.

1.0.14 Example

We define the category of sets with partial functions Set_p as follows:

(1) Objects are sets,

(2) Morphisms $X \to Y$ are partial functions from X to Y, which we view as pairs (U, f) where $U \subseteq X$ and $f\colon U \to Y$,

(3) Composition is defined as follows: for $(U, f)\colon X \to Y$ and $(V, g)\colon Y \to Z$, $(V, g) \circ (U, f) = (f^{-1}(V), g \circ f)$.

This is equivalent to Set_*, the category of pointed sets (objects are (X, x) for $x \in X$ and morphisms $f\colon (X, x) \to (Y, y)$ are functions $X \to Y$ such that $f(x) = y$). Indeed, we define a functor $F\colon \mathsf{Set}_p \to \mathsf{Set}_*$ by $F(X) = (X \cup \mathsf{Set}*, *)$ and for $(U, f)\colon X \to Y$ define $F(U, f)\colon X \cup \mathsf{Set}* \to Y \cup \mathsf{Set}*$ by $* \mapsto *$, $x \mapsto f(x)$ if defined and otherwise to $*$. This is essentially surjective clearly, and fully faithful clearly.

1.0.15 Definition

For a category $\mathcal{C}$, we define its **center** to be the set of natural transformations $1_{\mathcal{C}} \Rightarrow 1_{\mathcal{C}}$. Denote this by $Z(\mathcal{C})$.

Note that $Z(\mathcal{C})$ equipped with composition (vertical and horizontal agree on the center) is a commutative monoid. Clearly it is a monoid, as the natural transformation defined by the identity on all components is its identity. And it is commutative; by naturality for every $f\colon X \to X$ one has $f\alpha_X = \alpha_X f$, in particular for $f = \beta_X$.

1.0.16 Proposition

If $\mathcal{C}$ and $\mathcal{D}$ are equivalent categories, then their centers are isomorphic commutative monoids.

Proof

Let $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon G$ be an equivalence, with isomorphisms $\epsilon\colon FG \cong 1_D$ and $\eta\colon 1_C \cong GF$. Then we define $Z(\mathcal{C}) \to Z(\mathcal{D})$ as follows: $\alpha \mapsto \epsilon \circ F\alpha G \circ \epsilon^{-1}$. This is a monoid morphism: $(\epsilon \circ F1G \circ \epsilon^{-1})_X = \epsilon_X \circ 1_{FGx} \circ \epsilon_X^{-1} = 1_X$ and since $F\alpha\beta G = (F\alpha G)(F\beta G)$ we get commutativity. Its inverse is given by $\beta \mapsto \eta^{-1} \circ G\beta F \circ \eta$. ■

2 Limits

2.1 First examples of (co)limits

Our first example of a limit is a **product**. Suppose we have objects $X, Y \in \mathcal{C}$, their product is an object $X \times Y$ with associated morphisms $p_X\colon X \times Y \to X$ and $p_Y\colon X \times Y \to Y$ such that for any other object $C \in \mathcal{C}$ with $q_X\colon C \to X$ and $q_Y\colon C \to Y$, there exists a unique $f\colon C \to X \times Y$ making the following diagram commute:

$$\begin{tikzcd}
 & & X \\
C \arrow[urr, "q_X"] \arrow[r, dashed, "f"] \arrow[drr, "q_Y"'] & X \times Y \arrow[ur, "p_X"] \arrow[dr, "p_Y"] & \\
 & & Y
\end{tikzcd}$$

Such an object $X \times Y$, if it exists, is unique up to isomorphism. This isomorphism is in fact the unique isomorphism which commutes (as above) with p_X, p_Y.

This construct can clearly be generalized to products of families of objects, $\{X_i\}_{i \in I}$. Here are some examples:

(1) For objects $\{x_i\}_{i \in I}$ in a partial order P (a partial order viewed as a category has a unique morphism $x \to y$ iff $x \leq y$), their product is their infimum (if it exists). Indeed, it must be an object $p \in P$ such that for every $c \leq x_i$, $c \leq p$.

(2) In Set, the product is the usual set-theoretic product, and the morphisms $p_i\colon \prod_{i \in I} X_i \to X_i$ are the projections.

(3) In $\mathsf{Grp}, \mathsf{Ab}, \mathsf{Ring}, \mathsf{Vec}_k$, etc. the product is the usual product of such algebraic structures (i.e. the set-theoretic product with the usual operations).

(4) In Top, the category of topological spaces, the product of topological spaces X_i is their set-theoretic product endowed with the *product topology*. This topology has a basis $\prod_i U_i$ for $U_i \subseteq X_i$ open, with all but finitely many being equal to X_i.

(5) In the category of metric spaces, the product of X_i is their set-theoretic product with the supremum metric $d((x_i)_i, (y_i)_i) = \sup_{i \in I} d(x_i, y_i)$. Of course this only exists if the metric spaces are uniformly bounded, or we allow for infinite distance.

(6) The product in the category of (small or locally small) categories is the usual product of categories.

Our first example of a colimit is a **coproduct**. It is the dual of a product: the coproduct of $X, Y \in \mathcal{C}$ is an object $X \coprod Y \in \mathcal{C}$ with morphisms $p_X\colon X \to X \coprod Y$ and $p_Y\colon Y \to X \coprod Y$ such that for every other object C with $q_X\colon X \to C$ and $q_Y\colon Y \to C$, there exists a unique $f\colon X \coprod Y \to C$ such that the dual diagram to the one above commutes. That is, $fp_x = q_X$ and $fp_Y = q_Y$. Equivalently, the coproduct of objects is their product in $\mathcal{C}^{\mathsf{op}}$.

(1) The coproduct of objects in a partial order is their supremum.

(2) The coproduct in Set is the *disjoint union*: $\coprod_{i \in I} X_i$, which is the union of the sets $X_i \times \mathsf{Set} i$.

(3) The coproduct in Ab is the direct sum: $\bigoplus_{i \in I} X_i$ the subset of $\prod_{i \in I} X_i$ for which all but finitely many elements are 0.

(4) Coproducts in Grp is the free product (not discussed here).

(5) Coproducts in Top is the set-theoretic disjoint union $\coprod_{i\in I} X_i$ whose topology has a base given by the open sets in each X_i.

(6) In the category of commutative rings, the coproduct of R, S is their tensor product $R \otimes S$ with multiplication given by $(r \otimes s) \cdot (r' \otimes s') = (rr) \otimes (ss')$.

2.2 In general

We can generalize the notion of products and coproducts to **limits** and **colimits**.

2.2.1 Definition

For a (small) category I, we define its **initial completion** and **terminal completion** as I_i and I_f. In both I_i and I_f we add a new object to I, call it $*$. In I_i we add a unique morphism $* \to x$ for every $x \in I$, and for I_f a unique morphism $x \to *$ for every $x \in I$ (this uniquely determines composition). Clearly $*$ is an initial object in I_i and final in I_f.

2.2.2 Definition

Given a (small) diagram (a functor) $F\colon I \to \mathcal{C}$, a **cone** over F is a functor $\hat{F}\colon I_f \to \mathcal{C}$ which agrees with F on I. A morphism of cones $\hat{F} \to \hat{F}'$ is a natural transformation $\alpha\colon \hat{F} \Rightarrow \hat{F}'$ such that $\alpha_i = \mathrm{id}_{Fi}$ for every $i \in I$. This forms a category of cones, $\mathcal{C}_{/F}$.

Dually a **cocone** under F is a functor $\hat{F}\colon I_i \to \mathcal{C}$ which agrees with F on I, and morphisms are defined similarly. This forms a category of cocones, $\mathcal{C}_{F/}$.

Thus a cone over F is simply an object $\hat{F}(*) = X$ and morphisms $\hat{F}(* \to i) = f_i\colon X \to X_i$ (where $Fi = X_i$) such that for every $\phi\colon i \to j$ in I, we have $F(\phi)s_i = s_j$. So we can view a cone simply as this data $(X, s_i)_{i\in I}$. A morphism of cones $(X, s_i) \to (Y, t_i)$ is simply a morphism $f\colon X \to Y$ such that $s_i = t_i f$ for every $i \in I$.

Dually a cocone can be viewed as $(X, s_i)_{i\in I}$ such that for every $\phi\colon i \to j$ in I, we have $s_j F(\phi) = s_i$. A morphism of cocones $(X, s_i) \to (Y, t_i)$ is simply a morphism $f\colon X \to Y$ such that $t_i = f s_i$.

2.2.3 Definition

A **limit** of F is a terminal object in $C_{/F}$, and a **colimit** is an initial object in $C_{F/}$.

Since terminal and initial objects are unique up to unique isomorphism, the limits of a functor (if they exist) are all isomorphic (as are colimits), with a unique isomorphism that commutes with the diagram.

We denote the (we can say "the" since it is unique up to isomorphism) limit of F by $\lim_I F$ and its colimit by $\mathrm{colim}_I F$. Formally these are objects of $\mathcal{C}$ along with extra information, but generally we keep that information implicit, and write $\lim_I F, \mathrm{colim}_I F \in \mathcal{C}$ to be their associated objects.

We note that a colimit is just a limit in the opposite category. Explicitly, a cocone under $F\colon I \to \mathcal{C}$ is the same as a cone over $F^{\mathsf{op}}\colon I^{\mathsf{op}} \to \mathcal{C}^{\mathsf{op}}$: $I_f^{\mathsf{op}} = (I_i)^{\mathsf{op}}$ and so a cone over F^{op} is a functor $\hat{F}^{\mathsf{op}}\colon (I_i)^{\mathsf{op}} \to \mathcal{C}^{\mathsf{op}}$ which agrees with F^{op} on I, equivalently a functor $\hat{F}\colon I_I \to \mathcal{C}$ which agrees with F on I. That is to say $\mathcal{C}^{\mathsf{op}}_{/F^{\mathsf{op}}} \cong \mathcal{C}_{F/}$.

2.3 More examples

Let us consider the index category $I = \bullet \rightrightarrows \bullet$ (two objects, two non-identity morphisms), and diagrams over it. Such a diagram is $X \overset{f}{\underset{g}{\rightrightarrows}} Y$. Explicitly a cone over this diagram is an object C and $h\colon C \to X, h'\colon C \to Y$ such that $fh = gh = h'$. Thus h' is redundant and we can view a cone as simple (C, h) where $h\colon C \to X$ such that $fh = gh$.

The limit of this diagram is called an **equalizer**: it is (E, h) with $h\colon E \to X$ such that $fh = gh$ and for every other cone (C, h') there is a unique morphism $\phi\colon C \to E$ such that $h' = h\phi$.

For example, in Set the equalizer is just $\{x \in X \mid fx = gx\} \subseteq X$. This is clearly a cone, and if $fh = gh$ for any other $h\colon C \to X$ we have that h restricts to being in the equalizer, so ϕ is just the inclusion.

2.4 Limits as products and equalizers

Every limit (in any category) can be written using products and equalizers. Indeed, let $F: I \to \mathcal{C}$ be a diagram and let us consider the following diagram:

$$f, g: \prod_{i \in I} F(i) \rightrightarrows \prod_{\alpha: i \to j \in I} F(j)$$

we construct f, g and show that the limit of F is the equalizer of the above diagram.

We begin by defining f. In general $\hom(\bullet, \prod_{i \in I} X_i) \cong \prod_{i \in I} \hom(\bullet, X_i)$ naturally, since $f: Y \to \prod_{i \in I} X_i$ corresponds to $f_i = p_i f$, and f_i gives us a unique f which factors through the f_is by the universal property of products. In particular, to define f we need to define $f_\alpha: \prod_{i \in I} F(i) \to F(j)$ for every $\alpha: i \to j$ to simply be p_j. And similarly we define $g_\alpha: \prod_{i \in I} F(i) \to F(j)$ to be the composition $F(\alpha) p_i$.

2.4.1 Theorem

If $\phi: X \to \prod_{i \in I} F(i)$ is an equalizer for the above diagram, then $(\phi_i: X \to F(i))$ is a limit to the diagram F.

Proof

Since ϕ is an equalizer for f, g, i.e. $f\phi = g\phi$ then for $\alpha: i \to j$ we compose with p_j to get $p_j \phi = F(\alpha) p_i \phi$, i.e. $\phi_j = F(\alpha)\phi_i$. This is exactly what it means for (ϕ_i) to be a cone for F though. So F-cones can be viewed as equalizer cones, and then equalizer morphisms are the same as F-cone morphisms. Indeed, if $\psi: Y \to \prod_{i \in I} F(i)$ is another equalizer cone, then a morphism $\psi \to \phi$ is a morphism $f: Y \to X$ such that $\psi = \phi f$. That is equivalent to $\psi_i = \phi_i f$ for all i, i.e. a morphism of F-cones. So we see that equalizers and F-limits are the same, as desired. ■

Note that the limit of F can exist without $\prod_{i \in I} F(i)$ or $\prod_{\alpha: i \to j} F(j)$ existing. But if a category has arbitrary products and equalizers, then it has all limits.

Similarly for cocones, by dualizing, we get that a coequalizer (since $I^{\mathsf{op}} = I$) of $X \rightrightarrows f[g]Y$ is an $h: Y \to E$ such that $hf = hg$, and is universal in this regard. And so given a diagram $F: I \to \mathcal{C}$, a limit is a coequalizer of

$$f, g: \coprod_{\alpha: i \to j} F(j) \to \coprod_{i \in I} F(i)$$

where $f_\alpha: F(j) \to \coprod_{i \in I} F(i)$ is the inclusion ι_j and $g_\alpha: F(j) \to \coprod_{i \in I} F(i)$ is the composition $F(\alpha)\iota_i$.

2.4.2 Definition

A category $\mathcal{C}$ is **complete** if it has all (small) limits. A category $\mathcal{C}$ is **cocomplete** if it has all (small) colimits.

2.4.3 Corollary

A category is complete iff it has arbitrary products and equalizers. A category is cocomplete iff it has arbitrary coproducts and coequalizers.

2.4.4 Example

In Set, we already have a characterization of products, coproducts, equalizers, and coequalizers. Thus the

limit of $F: I \to \mathsf{Set}$ is simply (since for $\alpha: i \to j$, $f_\alpha(x_i)_{i\in I} = x_j$ and $g_\alpha(x_i)_{i\in I} = F(\alpha)(x_i)$),

$$\lim_I F = \left\{ (x_i)_{i\in I} \in \prod_{i\in I} F(i) \;\middle|\; \forall \alpha: i \to j.\, x_j = F(\alpha)(x_i) \right\}$$

The coequalizer of f, g in Set is $Y/\sim$ where $\sim$ is the smalleste equivalence relation such that $f(x) \sim g(x)$ for all $x \in X$. Thus

$$\operatorname{colim}_I F = \coprod_{i\in I} F(i)/(\forall \alpha: i \to j.\, x_j \sim F(\alpha)x_i)$$

Since we already know that the following categories have products and coproducts, and we can find equalizers and coequalizers:

2.4.5 Corollary

$\mathsf{Set}, \mathsf{Ab}, \mathsf{Top}, \mathsf{Grp}, \mathsf{Ring}, \mathsf{CRing}, \mathsf{Vec}_k$ are all complete and cocomplete.

Note that the restriction of (co)complete categories to *small* limits and colimits is necessary:

2.4.6 Proposition

If $\mathcal{C}$ is a category such that every diagram (not just small ones) has a limit, then $\mathcal{C}$ is a preorder (a partial order which is not necessarily antisymmetric).

Proof

We need to show that between every two objects there is at most a single morphism. Suppose that there are two morphisms $f, g: X \rightrightarrows Y$, then let κ be the cardinality of all the morphisms in $\mathcal{C}$. Consider then Y^κ (the product $\prod_\kappa Y$), then since $\hom(X, Y^\kappa) \cong \hom(X, Y)^\kappa$, we have that there are at least 2^κ morphisms $X \to Y^\kappa$, a contradiction since $2^\kappa > \kappa$. ■

2.4.7 Proposition

If I has an initial element i_0, then for every $F: I \to \mathcal{C}$, $\lim_I F = F(i_0)$. Dually if I has a terminal element, then $\operatorname{colim}_I F = F(i_0)$.

Proof

A cone over F is $(X, s_i: X \to X_i)$ such that $F(\phi)s_i = s_j$ for all $\phi: i \to j$. Let $\phi: i_0 \to i$ be the unique morphism, then we have $s_i = F(\phi)s_{i_0}$ and so we can view a cone simply as $(X, s_{i_0}: X \to X_{i_0})$. Thus X_{i_0} is a cone ($s_{i_0} = \mathrm{id}$) and it clearly is terminal, thus it is a limit. ■

2.4.8 Example

A **directed set** is a partial order $(I, \leq)$ such that every finite set has an upper bound; equivalently for every $a, b \in I$ there exists a $c \in I$ such that $a, b \leq c$. The colimit of a functor $I \to \mathcal{C}$ is called a **direct limit**, while the limit of a functor $I^{\mathsf{op}} \to \mathcal{C}$ is called a **inverse limit**.

A functor $I \to \mathcal{C}$ is also called a **direct system**, it can be characterized as a collection of objects $\{X_i\}_{i\in I}$ and morphisms $\{f_{ij}\colon X_i \to X_j\}_{i\le j}$ such that $f_{ii} = \mathrm{id}_{X_i}$ and $f_{ik} = f_{jk} \circ f_{ij}$. In algebraic categories like $\mathsf{Grp}, \mathsf{Ring}$, etc. we can characterize the direct limit as

$$\operatorname{colim} X_i = \bigsqcup_{i\in X_i} X_i / \{x_i \sim x_j \iff \exists i, j \le k.\, f_{ik}(x_i) = f_{jk}(x_j)\}$$

where $\sqcup$ is the set-theoretic disjoint union.

A functor $I^{\mathsf{op}} \to \mathcal{C}$ is also called a **inverse system**, a collection of objects $\{X_i\}_{i\in I}$ and morphisms $\{f_{ij}\colon X_j \to X_i\}_{i\le j}$ such that $f_{ii} = \mathrm{id}_{X_i}$ and $f_{ik} = f_{ij} \circ f_{jk}$ for $i \le j \le k$. In algebraic categories the inverse limit is

$$\lim X_i = \left\{ (x_i)_{i\in I} \in \prod_{i\in I} X_i \;\middle|\; x_i = f_{ij}(x_j) \text{ for all } i \le j \right\}$$

In particular if $f_{ij}\colon X_i \hookrightarrow X_j$ are inclusions then $\operatorname{colim} X_i$ is simply the union of the X_is. And if $f_{ij}\colon X_j \hookrightarrow X_i$ are inclusions then $\lim X_i$ is the intersection.

We consider now the directed set ω (finite numbers), a functor $\omega \to \mathcal{C}$ is characterized simply by morphisms $X_i \to X_{i+1}$. More interestingly consider the maps $\mathbb{Z} \xrightarrow{\times p} \mathbb{Z}$ for prime p. The direct limit of this direct system $(\mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \cdots)$ is $\mathbb{Z}[1/p] \subseteq \mathbb{Q}$, the subgroup of all rational numbers whose denominator is a power of p. The projections $\mathbb{Z}_i \to \mathbb{Z}[1/p]$ (where $\mathbb{Z}_i$ is the ith copy of $\mathbb{Z}$ in the system) are given by $\times 1/p^i$.

An inverse system of ω are morphisms $X_{i+1} \to X_i$. Let us consider the inverse system whose morphisms $\mathbb{Z}/p^{k+1}\mathbb{Z} \to \mathbb{Z}/p^k\mathbb{Z}$ are inclusion. The inverse limit of this inverse system is called $\mathbb{Z}_p$, the **p-adic integers**, which are $(x_n)_n$ such that $x_n \in \mathbb{Z}/p^n\mathbb{Z}$ and $x_n \equiv x_{n+1} \pmod{p^n}$.

Another example of an interesting inverse limit is the inverse limit of the ω^{op}-system of morphisms $S^1 \to S^1$ given by wrapping the circle p times around itself. This gives an interesting compact Hausdorff topological space.

2.5 Group actions

A group G can be viewed as a category $\mathsf{B}G$ with a single object whose hom-set is G. Then a G-set is just a functor $F\colon \mathsf{B}G \to \mathsf{Set}$ (i.e. the two categories are isomorphic). Indeed, a functor $F\colon \mathsf{B}G \to \mathsf{Set}$ is a set X along with automorphisms f_g for every $g \in G$ which preserve composition.

Since $\mathsf{B}G$ is a small category, we can consider the limits and colimits of G-actions. These are the invariants and coinvariants of the action, respectively:

$$\lim F = \{x \in X \mid \forall g \in G.\, gx = x\} = X^G, \qquad \operatorname{colim} F = X / \{\forall g \in G.\, gx \sim x\} = X_G$$

If we swap Set with Top, then a functor $F\colon \mathsf{B}G \to \mathsf{Top}$ is just a topological space X and homeomorphisms for every $g \in G$ which preserve composition. The limit of such a functor is still the subspace of G-invariants, while the colimit is the space of paths with the quotient topology $X \to X/G$.

And if we swap Set with Vec_k, then a functor $F\colon \mathsf{B}G \to \mathsf{Vec}_k$ is a G-representation. The limit is still the subspace of G-invariants, and the colimit is $X/\operatorname{span}\{x - gx \mid g \in G, x \in X\}$.

2.6 Pullbacks and pushforwards

2.6.1 Definition

A **pullback** is the limit of a functor from the category $\bullet \to \bullet \leftarrow \bullet$. A **pushforward** is the colimit of a functor from the category $\bullet \leftarrow \bullet \to \bullet$ (the opposite category of that for pullbacks).

Since the category $\bullet \to \bullet \leftarrow \bullet$ has a terminal object, colimits over it are noninteresting, hence why we don't define "copullbacks". Similarly $\bullet \leftarrow \bullet \to \bullet$ has an initial object, so we don't define "copushforwards".

The limit cone of a pullback is also called a **pullback square**. That is, a pullback square is

$$\begin{tikzcd} P \arrow[r] \arrow[d] & Y \arrow[d, "g"] \\ X \arrow[r, "f"] & Z \end{tikzcd}$$

where P is the pullback of the diagram. In Set, the pullback of $X \xrightarrow{f} Z \xleftarrow{g} Y$ is $X \times_Z Y = \{(x, y) \in X \times Y \mid f(x) = g(y)\}$. ∎
Similarly the colimit cone of a pushforward is called a **pushforward square**:

$$\begin{tikzcd} Z \arrow[r, "g"] \arrow[d, "f"] & Y \arrow[d] \\ X \arrow[r] & P \end{tikzcd}$$

In Set the pushforward of $X \xleftarrow{f} Z \xrightarrow{g} Y$ is $X \cup_Z Y = X \uplus Y/(f(z) \sim g(z))$.

2.6.2 Proposition (Gluing lemma)

Given the following commutative diagram:

$$\begin{tikzcd} A \arrow[r, "p"] \arrow[d, "f"] & B \arrow[r, "q"] \arrow[d, "g"] & C \arrow[d, "h"] \\ A' \arrow[r, "p'"] & B' \arrow[r, "q'"] & C' \end{tikzcd}$$

(1) If the right square is a pullback square, then the left square is a pullback square iff the outer square is a pullback square.

(2) If the left square is a pushforward square, then the right square is a pushforward square iff the outer square is a pushforward square.

2.6.3 Lemma

If $\mathcal{C}$ has a terminal object 1, then the pullback of $A \to 1 \leftarrow B$ defines the product $A \times B$. Dually if $\mathcal{C}$ has an initial object 0, then the pushforward of $A \leftarrow 0 \to B$ defines the coproduct $A \sqcup B$.

Proof

A cone over $A \to 1 \leftarrow B$ is the data $(X, p_1 \colon X \to A, p_2 \colon X \to B, !\colon X \to 1)$ which commute. But since 1 is terminal, this is equivalent to $(X, p_1 \colon X \to A, p_2 \colon X \to B)$, i.e. a cone over A, B. Thus pullbacks of this diagram are simply products. ∎

2.6.4 Lemma

If $\mathcal{C}$ has binary products then the pullback of $A \xrightarrow{(f,g)} B \times B \xleftarrow{(1_B, 1_B)} B$ is the equalizer of $f, g \colon A \rightrightarrows B$. Dually for pushforwards and binary coproducts.

For $f \colon X \to A$ and $g \colon X \to B$ we denote by (f, g) the unique morphism $(f, g) \colon X \to A \times B$ which projects onto f and g.

Proof

A cone over $A \xrightarrow{(f,g)} B \times B \xleftarrow{(1_B,1_B)} B$ is the data $(X, \phi\colon X \to A, \psi\colon X \to B, \chi\colon X \to B \times B)$ such that

$$(f,g) \circ \phi = (1_B, 1_B) \circ \psi = \chi$$

so equivalently (X, ϕ, ψ) such that $(f,g)\phi = (1_B, 1_B)\psi$, which is equivalent to requiring that $f\phi = \psi, g\phi = \psi$, so equivalently a cone is just $(X, \phi\colon X \to A)$ such that $f\phi = g\phi$, which is a cone over the equalizer diagram of f, g. ■

By considering our proof from before relating products and equalizers to all (small) limits, we see that we can write all finite limits with finite products and equalizers. Similar for colimits. Further note that the finite product $X_1 \times \cdots \times X_n$ is equal to $X_1 \times (X_2 \times \cdots)$ (if all binary products exist). In general if $\mathcal{C}$ has all products up to a certain size, then for a set I and a partition of I, $\{I_j\}_{j \in J}$ we have that $\prod_{j \in J} \prod_{i \in I_j} X_i$ is the product $\prod_{i \in I} X_i$. We will omit the proof of that here, but it means that a category has all finite products iff it has all binary products. Similar for coproducts.

Furthermore, consider the empty category 0. A functor $F\colon 0 \to \mathcal{C}$ has a limit iff $\mathcal{C}$ has a terminal object. Indeed, $\mathcal{C}_{/F}$ is just $\mathcal{C}$, so a limit of F (a terminal cone) is just a terminal object in $\mathcal{C}$. Since 0 is a discrete category (i.e. no non-trivial morphisms), we can view a terminal object as the empty product. Similarly an initial object is the empty coproduct.

2.6.5 Corollary

A category has all finite limits iff it has all pullbacks and a terminal object. Dually a category has all finite colimits iff it has all pushforwards and an initial object.

2.7 Preservation of (co)limits

2.7.1 Definition

Let I be an index category, and $c \in \mathcal{C}$ an object. We define the **constant functor** at c to be the functor $c\colon I \to \mathcal{C}$ which maps all objects in I to c and all morphisms to 1_c. This defines the so-called **diagonal functor** $\Delta\colon \mathcal{C} \to \mathcal{C}^I$ which maps $c \in \mathcal{C}$ to the constant functor at c, and $f\colon c \to d$ to be the natural transformation whose components at each $i \in I$ to be f.

Recall that a cone over $F\colon I \to \mathcal{C}$ is a functor $\hat{F}\colon I_f \to \mathcal{C}$ which agrees with F. That is, it is the data $(c, f_i\colon c \to F(i))_{i \in I}$ such that for every $\phi\colon i \to j \in I$ we have that $F(\phi)f_i = f_j$. Notice that this is equivalent to a natural transformation $\lambda\colon c \Rightarrow F$ (from the constant functor to F): $\lambda_i = f_i$. So we can view cones over F to simply be natural transformations $\lambda\colon c \Rightarrow F$. A morphism of cones from $\lambda\colon c \Rightarrow F$ to $\mu\colon d \Rightarrow F$ is a $\mathcal{C}$-morphism $f\colon c \to d$ such that $\mu f = \lambda$. Dually we can consider cocones under F to be natural transformations $\lambda\colon F \Rightarrow c$.

Consider a diagram $F\colon I \to \mathcal{C}$ and a functor $G\colon \mathcal{C} \to \mathcal{D}$. This induces a diagram $G \circ F\colon I \to \mathcal{D}$, and as such we can try and relate $\lim_I G \circ F$ and $G(\lim_I F)$.

First notice that $G\colon \mathcal{C} \to \mathcal{D}$ induces a functor $G\colon \mathcal{C}_{/F} \to \mathcal{D}_{/F}$ by whiskering: $\lambda\colon c \Rightarrow F$ maps to $G\lambda\colon Gc \Rightarrow G \circ F$. Similarly it induces a functor $G\colon \mathcal{C}_{F/} \to \mathcal{D}_{F/}$ by whiskering as well: $\lambda\colon F \Rightarrow c$ maps to $G\lambda\colon G \circ F \Rightarrow Gc$. This leads us to the following definition:

2.7.2 Definition

Let $K\colon I \to \mathcal{C}$ be a class of diagram (i.e. a collection of functors $I \to \mathcal{C}$) and $G\colon \mathcal{C} \to \mathcal{D}$ be a functor. Then

(1) G **preserves** K's limits if λ is a limit cone of K (of some functor in K), $G\lambda$ is a limit cone of $G \circ K$.

(2) G **reflects** K's limits if for every for every cone $\mu \in \mathcal{C}_{/K}$, if $G\mu$ is a limit cone of $G \circ K$, then μ is

a limit cone of K.

(3) G **creates** K's limits if it reflects limits and if $G \circ K$ has a limit then there is a limit λ of K such that $G\lambda$ is a limit.

This dualizes to colimits.

So if $F: \mathcal{C} \to \mathcal{D}$ creates limits for some class of diagrams and $\mathcal{D}$ has limits of those diagrams, then $\mathcal{C}$ has all those limits and F preserves them.
As discussed, $G: \mathcal{C} \to \mathcal{D}$ induces a functor $G: \mathcal{C}_{/F} \to \mathcal{D}_{/F}$, so if $\lim_I G \circ F$ exists then it is terminal in $\mathcal{D}_{/F}$ and since $G(\lim_I F) \in \mathcal{D}_{/F}$ we have a unique morphism $\alpha: G(\lim_I F) \to \lim_I(G \circ F)$ (it is unique in $\mathcal{D}_{/F}$). Similarly we have a unique morphism $\beta: \operatorname{colim}_I(G \circ F) \to G(\operatorname{colim}_I F)$. These are called the **assembly maps**.
The following is immediate:

2.7.3 Proposition

G preserves F's limit (colimit) iff α (respectively β) is an isomorphism.

2.7.4 Example

Consider the **forgetful functor** $U: \mathsf{Ab} \to \mathsf{Set}$ which maps Abelian groups to their underlying sets and morphisms to their underlying pure functions. Given two Abelian groups A, B we have projections $A \xleftarrow{p_A} A \times B \xrightarrow{p_B} B$. Applying U gives us the diagram $U(A) \xleftarrow{U(p_A)} U(A \times B) \xrightarrow{U(p_B)} U(B)$, which is precisely a cone over $\{U(A), U(B)\}$. Thus we have a unique morphism (unique as a cone morphism):

$$U(A \times B) \to U(A) \times U(B)$$

This is an isomorphism, and as such U preserves binary products.
Similarly we have the coproduct $A \xrightarrow{j_A} A \oplus B \xleftarrow{j_B} B$, which gives us $U(A) \xrightarrow{j_A} U(A \oplus B) \xleftarrow{j_B} U(B)$ and thus we have a morphism

$$U(A) \sqcup U(B) \to U(A \oplus B)$$

which is not an isomorphism. Thus U does not preserve coproducts, and so U doesn't preserve colimits (U may preserve specific colimits though).

The following are immediate:

2.7.5 Proposition

Let $\mathcal{C}, \mathcal{D}$ be complete categories, then $F: \mathcal{C} \to \mathcal{D}$ preserves all limits iff it preserves all products and equalizers. Dually for colimits.

2.7.6 Lemma

If $G: \mathcal{C} \to \mathcal{D}$ and $H: \mathcal{D} \to \mathcal{E}$ are functors which preserve all I-limits (limits whose diagram is of shape I), then $H \circ G$ preserves all I-limits as well. Similar for colimits.

Notice that we can consider the composition for $F: I \to \mathcal{C}$:

$$H \circ G(\lim_I F) \to H(\lim_I G \circ F) \to \lim_I(H \circ G \circ F)$$

This is the assembly morphism for $H \circ G$. As such if $H \circ G$ preserves limits and H reflects isomorphisms (i.e. Hf is an isomorphism iff f is), then G preserves limits.

2.7.7 Example

If we consider the forgetful functors $\mathsf{Ab} \xrightarrow{U_1} \mathsf{Grp} \xrightarrow{U_2} \mathsf{Set}$, then since U_2 reflects isomorphisms and $U_2 \circ U_1$ preserves limits, U_1 preserves limits as well.

2.7.8 Proposition

Fully faithful functors reflect limits and colimits.

Proof

Let $F: I \to \mathcal{C}$ be a diagram and $G: \mathcal{C} \to \mathcal{D}$ be fully faithful. We claim that the induced functor $G: \mathcal{C}_{/F} \to \mathcal{D}_{/G\circ F}$ is fully faithful as well. Indeed, let $\lambda, \mu \in \mathcal{C}_{/F}$ be cones of objects c, d respectively; then $f \in \hom(G\lambda, G\mu)$ is a morphism $Gc \to Gd$ such that $(G\mu)f = G\lambda$. Clearly $G: \hom(\lambda, \mu) \to \hom(G\lambda, G\mu)$ is injective (since it is a restriction of $\hom(c, d) \to \hom(Gc, Gd)$). And since G is full, there is a $g: c \to d$ such that $Gg = f$; we claim that g is a morphism $\lambda \to \mu$. Indeed: we have $(G\mu)(Gg) = G(\mu g) = G\lambda$, and since G is fully faithful this means $\mu g = \lambda$.

This means that if $G\lambda$ is a limit cone, then for every $\mu \in \mathcal{C}_{/F}$ we have $\hom(\mu, \lambda) \cong \hom(G\mu, G\lambda) = \mathsf{Set}*$. Thus λ is terminal and thus a limit cone. So G reflects limits. ■

2.7.9 Corollary

If $F: \mathcal{C} \to \mathcal{D}$ is an equivalence, it preserves and creates all limits and colimits.

Proof

We claim that for any diagram $D: I \to \mathcal{C}$, $F: \mathcal{C}_{/D} \to \mathcal{D}_{/F\circ D}$ is an equivalence. We just need to show essential surjectivity, let $\lambda: d \Rightarrow F \circ D$ be a cone. Since F is an equivalence, $\iota: d \cong Fc$ for some $c \in \mathcal{C}$ and for each $i \in I$ there exists $\mu_i: c \to D(i)$ such that $\lambda_i = F(\mu_i)\iota$. Then μ is a cone $c \Rightarrow D$ such that $(F\mu)\iota = \lambda$, i.e. $F\mu$ and λ are isomorphic.

Since F is fully faithful we already know it reflects limits. Now if λ is a limit cone over $F \circ D$, then it is isomorphic to $F\mu$ so μ is a limit cone over D. Thus F creates all limits.

And if λ is a limit cone over D, then for every limit cone μ over $F\circ D$ it is isomorphic to $F\mu'$ so, $\hom(\mu, F\lambda) \cong \hom(F\mu', F\lambda) \cong \hom(\mu', \lambda)$ a singleton. Thus $F\lambda$ is a limit cone over $F \circ D$, so F preserves limits. ■

2.7.10 Corollary

Equivalent categories have the same shape limits.

2.7.11 Example

We claim that Vec_k and $\mathsf{Vec}_k^{\mathsf{op}}$ are not equivalent. Note that all vector spaces are isomorphic to coproducts of k. Now suppose that $E: \mathsf{Vec}_k \to \mathsf{Vec}_k^{\mathsf{op}}$ is an equivalence, then since it preserves coproducts it must map $\coprod_{i\in I} k$ to $\coprod_{i\in I} Ek$. But notice that coproducts in the opposite category are simply products, so $\coprod_{i\in I} k$ maps to $\prod_{i\in I} Ek$. Now notice that $k \cong \hom(k, k) \cong \hom(Ek, Ek)$, since in general V can be embedded into $\hom(V, V)$ we have that Ek has cardinality at most $|k|$.

Take I, J index sets, then we have that

$$\hom(\coprod_I k, \coprod_J k) \cong \prod_I \hom(k, \coprod_J k) \cong \prod_I \coprod_J k$$

and

$$\hom(\prod_J Ek, \prod_I Ek) \cong \prod_I \hom(\prod_J Ek, Ek)$$

in general we have that $\hom(V, Ek) \supseteq V$ (comparing cardinalities), and so we get that the above is a superset of $\prod_I \prod_J Ek \supset \prod_{I,J} k$. Thus by equivalence we must have that $\left|\prod_{I,J} k\right| \leq |\prod_I \coprod_J k|$. But for I, J large enough this means $\left|k^{I\times J}\right| \leq |J|^{|I|}$. Letting $J = k^I$ we get $k^J = k^{k^I} \leq k^I$ which is impossible.

2.7.12 Definition

An object $c \in \mathcal{C}$ can be viewed as a functor from the one-object category $c\colon 1 \to \mathcal{C}$ (there is an isomorphism $\mathcal{C} \cong \mathcal{C}^1$). The category of cones over c is called the **overcategory** of c and is denoted $\mathcal{C}/c$ (instead of $\mathcal{C}_{/c}$). Dually the category of cocones under c is called the **undercategory** of c and is denoted $c/\mathcal{C}$ (instead of $\mathcal{C}_{c/}$).

Now, a cone over c is just a natural transformation $\lambda\colon X \Rightarrow c$ for some $X \in \mathcal{C}$. Such a natural transformation is just a morphism $f\colon X \to c$. A morphism of cones from $f\colon X \to c$ to $g\colon Y \to c$ is a morphism $h\colon X \to Y$ such that $f = gh$. So we have an alternative characterization of $\mathcal{C}/c$:

(1) Objects are pairs (X, f) where $X \in \mathcal{C}$ and $f\colon X \to c$ is a morphism;

(2) A morphism $(X, f) \to (Y, g)$ is a morphism $h\colon X \to Y$ such that $f = gh$.

Similarly a cocone under c is a morphism $f\colon c \to X$. A morphism of cocones $f\colon c \to X$ to $g\colon c \to Y$ is a morphism $h\colon X \to Y$ such that $hf = g$.

In general we have a forgetful functor $U\colon \mathcal{C}_{/F} \to \mathcal{C}$ which maps $\lambda\colon c \Rightarrow F$ to c. And we also have $U\colon \mathcal{C}_{F/} \to \mathcal{C}$ which maps $\lambda\colon F \Rightarrow c$ to c. Special cases of these are of course forgetful functors from the overcategory and undercategory.

We can characterize the category of cones another way:

2.7.13 Definition

Let $F\colon \mathcal{C} \to \mathsf{Set}$ be a Set-valued functor. Then F's **category of elements** of $\mathcal{C}$ is the category $\int F$ whose objects are pairs (c, x) where $c \in \mathcal{C}$ and $x \in F(c)$, and a morphism $(c, x) \to (d, y)$ is a $\mathcal{C}$-morphism $f\colon c \to d$ such that $y = F(f)(x)$.

For a contravariant functor $F\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$, it has a category of elements of $\mathcal{C}^{\mathsf{op}}$. We define its category of elements of $\mathcal{C}$ to be the opposite category of this: $\int_{\mathcal{C}} F = (\int_{\mathcal{C}^{\mathsf{op}}} F)^{\mathsf{op}}$.

Explicitly, for a contravariant functor $F\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ its category of elements $\int_{\mathcal{C}} F$ has pairs (c, x) for $c \in \mathcal{C}$ and $x \in F(c)$ and a morphism $(c, x) \to (d, y)$ is a $\mathcal{C}$-morphism $f\colon c \to d$ such that $x = F(f)(y)$.

Let $F\colon I \to \mathcal{C}$ be a diagram, we define two functors:

$$\mathrm{Cone}(\bullet, F)\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}, \qquad \mathrm{Cone}(F, \bullet)\colon \mathcal{C} \to \mathsf{Set}$$

where $\mathrm{Cone}(c, F)$ is equal to the set of natural transformations $c \Rightarrow F$, and $\mathrm{Cone}(F, c)$ is the set of natural transformations $F \Rightarrow c$. For a morphism $f\colon c \to d$, $\mathrm{Cone}(f, F)\colon \mathrm{Cone}(d, F) \to \mathrm{Cone}(c, F)$ by $\lambda \mapsto \lambda \circ f$. Similarly $\mathrm{Cone}(F, f)\colon \mathrm{Cone}(F, c) \to \mathrm{Cone}(F, d)$ by $\lambda \mapsto f \circ \lambda$.

Now, we can characterize the category of cones and cocones as follows:

$$\mathcal{C}_{/F} = \int \mathrm{Cone}(\bullet, F), \qquad \mathcal{C}_{F/} = \int \mathrm{Cone}(F, \bullet)$$

Indeed, an object in the category of elements of $\mathrm{Cone}(\bullet, F)$ is a pair $(c, \lambda\colon c \Rightarrow F)$ and morphisms $(c, \lambda) \to (d, \mu)$ are morphisms $f\colon c \to d$ such that $\lambda = \mathrm{Cone}(f, F)(\mu) = \mu f$. This is precisely the definition of the category of cones. Similarly for cocones.

There is a clear forgetful functor $U\colon \int F \to \mathcal{C}$ which maps $(c, x) \mapsto c$. Note that because of our definition for contravariant functors, this is a forgetful functor both when F is covariant and when it is contravariant.

Let us now consider limits of the category of elements. To do so let us consider what it means for $D\colon I \to \int F$ to be a diagram. We can view such diagrams equivalently as diagrams: $UD\colon I \to \mathcal{C}$ as well as a cone $\lambda\colon * \Rightarrow FUD$ (viewing both as functors $I \to \mathsf{Set}$, where $*$ is a singleton). Indeed: if $D(i) = (c_i, x_i \in F(c_i))$ we can define λ to be $\lambda_i = x_i \in FUD(i)$. And conversely given a diagram $D\colon I \to \mathcal{C}$ and a cone $\lambda\colon * \to FD$ then this defines a diagram $I \to \int F$ by $i \mapsto (D(i), \lambda_i)$ and $\phi\colon i \to j$ maps to $D(\phi)$.

So a diagram $I \to \int F$ is simply a diagram $D\colon I \to \mathcal{C}$ as well as a cone $a\colon * \Rightarrow FD$ in Set. A morphism from a cone $\mu\colon (d, y) \Rightarrow (D, a)$ to $\lambda\colon (c, x) \Rightarrow (D, a)$ is a morphism $f\colon d \to c$ such that $x = Ff(y)$ and $U\mu = (U\lambda)f$.

2.7.14 Proposition

For a covariant functor F, the forgetful functor $U\colon \int F \to \mathcal{C}$ creates all limits which $\mathcal{C}$ admits and F preserves. What this means is that for every diagram $D\colon I \to \mathcal{C}$ which has a limit in $\mathcal{C}$ preserved by F, then this limit in $\mathcal{C}$ is created by U.

Proof

We need to show that if $D\colon I \to \mathcal{C}$ has a limit preserved by F, and $a\colon * \Rightarrow FD$ is a cone, then:

(1) if λ is a cone over (D, a) such that $U\lambda$ is a limit of D, then λ is a limit of (D, a).

(2) there exists a limit λ over (D, a) such that $U\lambda$ is a limit of D.

Let $\eta\colon \lim D \Rightarrow D$ be the limit cone of D, then since F preserves it $F\eta\colon F(\lim D) \Rightarrow FD$ is a limit cone as well.

Now suppose there is a $\bar{\eta}\colon X \Rightarrow (D, a)$ such that $U\bar{\eta} = \eta$. This means that $UX = \lim D$ and $\bar{\eta}_i = \eta_i$ for all $i \in I$. Thus, $\bar{\eta}\colon (\lim D, x) \Rightarrow (D, a)$ for $x \in F(\lim D)$, and since it is a cone we must have that $F(\eta_i)(x) = a_i$ for all $i \in I$ as well. Now we know that $F(\eta)\colon F(\lim D) \Rightarrow FD$ is a limit cone, and so there exists a unique morphism from $a\colon * \Rightarrow FD$ to $F(\eta)$. A morphism from a to $F(\eta)$ is precisely a choice of element $x \in F(\lim D)$ such that $F(\eta_i)(x) = a_i$ for all $i \in I$, as desired.

So there does indeed exist such a $\bar{\eta}\colon (\lim D, x) \Rightarrow (D, a)$ such that $U\bar{\eta} = \eta$. We claim that $\bar{\eta}$ is a limit of (D, a), proving both points. Let $\lambda\colon (c, y) \Rightarrow (D, a)$ be a cone, then $\hom(\lambda, \bar{\eta})$ is the set of all morphisms $f\colon c \to \lim D$ such that $x = Ff(y)$ and $\eta = (U\lambda)f$. This is a subset of $\hom(U\lambda, \eta)$ which is a singleton. Now, the extension of $U\lambda$ to λ corresponds to the morphism $a \to F(U\lambda)$ choosing y (as before), and the composition of this with the unique morphism $Ff\colon F(U\lambda) \to F(\eta)$ must be equal to the unique morphism $a \to F(\eta)$ corresponding to choosing x. This means that we have $Ff(y) = x$ as desired. So we have that $\bar{\eta}$ must indeed be terminal, as desired. ■

2.7.15 Corollary

If $\mathcal{C}$ is complete, then $U\colon \int F \to \mathcal{C}$ creates and preserves all limits F preserves.

If F is contravariant then the forgetful functor $U\colon \int_{\mathcal{C}} F \to \mathcal{C}$ is simply the opposite functor of $U^{\mathsf{op}}\colon \int_{\mathcal{C}^{\mathsf{op}}} F \to \mathcal{C}^{\mathsf{op}}$ which is the forgetful functor discussed above. Since U^{op} creates limits, this means that U creates colimits.

2.7.16 Corollary

If $F\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ is a contravariant functor then the forgetful functor $U\colon \int_{\mathcal{C}} F \to \mathcal{C}$ creates all colimits which $\mathcal{C}$ admits and F preserves.

In particular for our cone categories, we must consider what limits and colimits the functors $\mathrm{Cone}(F, \bullet)\colon \mathcal{C} \to \mathsf{Set}$ and $\mathrm{Cone}(\bullet, F)\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ preserve.

2.7.17 Proposition

If $\mathcal{C}$ is complete, then $U\colon \mathcal{C}_{F/} \to \mathcal{C}$ creates and preserves all limits. If $\mathcal{C}$ is cocomplete then $U\colon \mathcal{C}_{/F} \to \mathcal{C}$ creates and preserves all colimits.

Proof

By the above arguments, we just need to show that $\mathrm{Cone}(F, \bullet)$ and $\mathrm{Cone}(\bullet, F)$ preserve all limits and colimits respectively. Notice that if $\{A_i\}_{i\in I}$ are objects in $\mathcal{C}$ then $\mathrm{Cone}(F, \prod_{i\in I} A_i) \cong \prod_{i\in I} \mathrm{Cone}(F, A_i)$ naturally, and we also have that the cocone of an equalizer is naturally isomorphic to the equalizer of cocones. Thus $\mathrm{Cone}(F, \bullet)$ preserves all limits, and thus by our above arguments U creates and preserves all limits. ■

2.8 Limits in functor categories

For categories $\mathcal{C}, \mathcal{D}$ we consider the functor category $\mathcal{D}^{\mathcal{C}}$ of functors $\mathcal{C} \to \mathcal{D}$ whose morphisms are natural transformations. We alternatively write it by $[\mathcal{C}, \mathcal{D}]$.

2.8.1 Lemma

For categories $\mathcal{C}, \mathcal{D}, \mathcal{E}$ there is an isomorphism of categories $[\mathcal{C}, [\mathcal{D}, \mathcal{E}]] \cong [\mathcal{C} \times \mathcal{D}, \mathcal{E}] \cong [\mathcal{D}, [\mathcal{C}, \mathcal{E}]]$.

Proof

We give an isomorphism $\Phi\colon [\mathcal{C}, [\mathcal{D}, \mathcal{E}]] \to [\mathcal{C} \times \mathcal{D}, \mathcal{E}]$. For a functor $F\colon \mathcal{C} \to \mathcal{E}^{\mathcal{D}}$, we define $\Phi(F)\colon \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ by $\Phi(F)(c, d) = F(c)(d)$ on objects and if $\alpha\colon F \Rightarrow G$ is a natural transformation, we note that this means for every $c \in \mathcal{C}$, $\alpha_c\colon Fc \Rightarrow Gc$ is a natural transformation of functors $\mathcal{D} \to \mathcal{E}$, and so it has a component $\alpha_{c,d}\colon F(c)(d) \to G(c)(d)$. So we define $\Phi(\alpha)\colon \Phi(F) \Rightarrow \Phi(G)$ is given by $\Phi(\alpha)_{(c,d)} = \alpha_{c,d}$. This is a functor.

We define its inverse $\Psi\colon [\mathcal{C} \times \mathcal{D}, \mathcal{E}] \to [\mathcal{C}, [\mathcal{D}, \mathcal{E}]]$ as follows: for $F\colon \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ define $\Psi(F)\colon \mathcal{C} \to \mathcal{E}^{\mathcal{D}}$ by $\Psi(F)(c)(d) = F(c, d)$ on objects $c \in \mathcal{C}$. For morphisms $f\colon c \to c'$ we define $\Psi(F)(f)\colon \Psi(F)(c) \Rightarrow \Psi(F)(c')$ by $\Psi(F)(f)_d = F(f, 1_d)$. And for $\alpha\colon F \Rightarrow G$ natural, we define $\Psi(\alpha)_{c,d}\colon F(c, d) \to G(c, d)$ to be $\alpha_{(c,d)}$.

The last isomorphism follows because $\mathcal{C} \times \mathcal{D} \cong \mathcal{D} \times \mathcal{C}$. ■

This gives us an isomorphism $[\mathcal{C}, [\mathcal{D}, \mathcal{E}]] \cong [\mathcal{D}, [\mathcal{C}, \mathcal{E}]]$ which maps $F\colon \mathcal{C} \to \mathcal{E}^{\mathcal{D}}$ to $\hat{F}\colon \mathcal{D} \to \mathcal{E}^{\mathcal{C}}$ defined by

(1) $\hat{F}(d)(c) = F(c)(d)$,

(2) for $f\colon c \to c'$, $\hat{F}(d)(f) = F(f)_d$,

(3) for $f\colon d \to d'$, $\hat{F}(f)_c = F(c)(f)$.

2.8.2 Theorem

Let $F\colon I \to [\mathcal{C}, \mathcal{D}]$ be a diagram and consider its curried functor $\hat{F}\colon \mathcal{C} \to [I, \mathcal{D}]$. Show that if for every $X \in \mathcal{C}$, $\hat{F}(X)\colon I \to \mathcal{D}$ has a limit $\lim_I \hat{F}(X)$, then F has a limit and there are natural isomorphisms

$$(\lim_I F)(X) \cong \lim_I \hat{F}(X)$$

Proof

We need to show that there is a functor $\lim_I F: \mathcal{C} \to \mathcal{D}$ which acts on objects by $(\lim_I F)(X) = \lim_I \hat{F}(X)$ and that it has an associated terminal cone. For brevity we write $L: \mathcal{C} \to \mathcal{D}$ instead of $\lim_I F$.

Let $f: X \to Y$ be a $\mathcal{C}$-morphism, then we have a natural transformation $\hat{F}(f): \hat{F}(X) \Rightarrow \hat{F}(Y)$. Now let $\lambda_X: \lim_I \hat{F}(X) \Rightarrow \hat{F}(X)$ be the limit cone, then $\hat{F}(f)\lambda_X: \lim_I \hat{F}(X) \Rightarrow \hat{F}(Y)$ is another cone, and as such there is a unique $\mathcal{D}$-morphism $L(f): \lim_I \hat{F}(X) \Rightarrow \lim_I \hat{F}(Y)$ such that $\lambda_Y L(f) = \hat{F}(f)\lambda_X$. So we define L by:

$$L(X) = \lim_I \hat{F}(X), \qquad \lambda_Y L(f) = \hat{F}(f)\lambda_X$$

This is functorial: $\hat{F}(gf)\lambda_X = \hat{F}(g)\hat{F}(f)\lambda_X = \hat{F}(g)\lambda_Y L(f) = \lambda_Z L(g)L(f)$, since $L(gf)$ is the unique morphism satisfying this, we have $L(gf) = L(g)L(f)$.

Now we need to find a cone $\Lambda: L \Rightarrow F$, that is for every $i \in I$ a natural transformation $\Lambda_i: L \Rightarrow F(i)$ such that everything commutes. For $X \in \mathcal{C}$ we have the projection $(\lambda_X)_i: \lim_I \hat{F}(X) = L(X) \to \hat{F}(X)(i) = F(i)(X)$, so define $\Lambda_{i,X} = \lambda_{X,i}$. Each Λ_i is natural: this assertion just means that $\Lambda_{iY}L(f) = F(i)(f)\Lambda_{iX}$ for all $f: X \to Y$. Expanding definitions, this just means $\lambda_{Yi}L(f) = \hat{F}(f)_i\lambda_{Xi}$, which clearly holds by definition of $L(f)$. And Λ as a whole is natural: to show $\Lambda_j = F(\phi)\Lambda_i$ for every $\phi: i \to j$ it is necessary and sufficient to show this for every component, that is to show $\lambda_{Xj} = F(\phi)_X\lambda_{Xi}$. This is true by naturality of λ_X (since $\hat{F}(X)(\phi) = F(\phi)_X$).

Now we claim that it is terminal. Let $f: \mathcal{C} \to \mathcal{D}$ be another functor with a cone $\Phi: f \Rightarrow F$, we need to show that there is a unique natural transformation $\alpha: f \Rightarrow F$ such that $\Phi = \Lambda\alpha$. This is equivalent to saying that for every $i \in I, X \in \mathcal{C}$ we have $\Phi_{iX} = \lambda_{Xi}\alpha_X$. And indeed this is true: $\Phi_{\bullet X}: f(X) \Rightarrow \hat{F}(X)$ is a cone over $\hat{F}(X)$ and since λ_X is a limit cone there exists a unique α_X such that $\Phi_{\bullet X} = \lambda_X\alpha_X$. This α is natural and thus concludes our proof. ■

This tells us that limits (and by duality colimits) in functor categories can be computed "pointwise": if $F: I \to [\mathcal{C}, \mathcal{D}]$ is a diagram then we have

$$(\lim_I F_i)(X) = \lim_I (F_i(X))$$

2.8.3 Corollary

If $\mathcal{D}$ has all (co)limits of shape I, then so does the functor category $[\mathcal{C}, \mathcal{D}]$. In particular the functor categories over (co)complete categories are all (co)complete.

2.9 Functorality of limits and colimits

Notice that if $\mathcal{C}$ has all limits of shape I, then we can define a functor

$$\lim_I: [I, \mathcal{C}] \to \mathcal{C}$$

which is defined for functors to be the limit of F (we need choice to choose an explicit representation of the limit in $\mathcal{C}$, as it is only defined up to isomorphism). For a natural transformation $\alpha: F \Rightarrow G$ of functors $I \to \mathcal{C}$, note that if $\lambda_F: \lim_I F \Rightarrow F$ is the limit cone over F, then $\alpha\lambda_F: \lim_I F \Rightarrow G$ is a cone as well, and as such there is a unique morphism $f_\alpha: \lim_I F \to \lim_I G$ such that $\alpha\lambda_F = \lambda_G f_\alpha$, so define $\lim_I(\alpha) = f_\alpha$. This is the same construction as before; it is functorial for the same reason.

Similarly if $\mathcal{C}$ has all colimits of shape I we define

$$\operatorname{colim}_I: [I, \mathcal{C}] \to \mathcal{C}$$

2.9.1 Theorem

If $F: I \times J \to \mathcal{C}$ is a diagram, then considering its curried functors $I \to [J, \mathcal{C}]$ and $J \to [I, \mathcal{C}]$ we have

$$\lim_{i \in I} \lim_{j \in J} F(i,j) \cong \lim_{I \times J} F \cong \lim_{j \in J} \lim_{i \in I} F(i,j)$$

when either the first or last limits exist (and their existence implies the existence of the middle limit).

We omit the proof for now, and will give a slick one in the future. We also have:

2.9.2 Theorem

$\lim_I$ preserves all limits and colim_I all colimits.

Which we will prove slickly in the future as well.

3 Yoneda's Lemma

3.1 Representable functors

3.1.1 Definition

Let $\mathcal{C}$ be a category, then we have a bifunctor

$$\hom\colon \mathcal{C}^{\mathsf{op}} \times \mathcal{C} \to \mathsf{Set}$$

call the **hom-functor** which maps pairs of objects c, d to their hom-set $\hom(c, d)$. Pairs of morphisms $f\colon c \to d$ and $f'\colon c' \to d'$ are mapped to the function $\hom(f, f')\colon \hom(d, c') \to \hom(c, d')$ mapping $g\colon d \to c'$ to $f' \circ g \circ f\colon c \to d'$. I.e. $\hom(f, f') = f \bullet f'$.

The hom-functors restrict along $\mathcal{C}^{\mathsf{op}}$ and $\mathcal{C}$. Specifically for any $c \in \mathcal{C}$ we have single-variable functors

$$\hom(c, \bullet)\colon \mathcal{C} \to \mathsf{Set}, \qquad \hom(\bullet, c)\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$$

where $\hom(c, f) = 1_C \bullet f$ which we will denote by f_* and $\hom(c, f) = f \bullet 1_C = f^*$.

3.1.2 Definition

A covariant functor $F\colon \mathcal{C} \to \mathsf{Set}$ is called **corepresentable** if there is some $X \in \mathcal{C}$ such that F is naturally isomorphic to the hom-functor $\hom(X, \bullet)$. A contravariant functor $F\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ is **representable** if F is naturally isomorphic to $\hom(\bullet, X)$ for some $X \in \mathcal{C}$. It is said that F is **(co)represented** by $X \in \mathcal{C}$.

We will sometimes say "represented" instead of "corepresented" when the variance is clear.

3.1.3 Example

(1) The forgetful functor $U\colon \mathsf{Grp} \to \mathsf{Set}$ is represented by $\mathbb{Z}$. That is, we have a natural isomorphism $UG \cong \hom(\mathbb{Z}, G)$ which maps $\phi\colon \mathbb{Z} \to G$ to $\phi(1)$. This is a bijection (because $\mathbb{Z}$ is the free group on a single generator), and it can be seen to be natural.

(2) In a somewhat generalization, the forgetful functor from the category of R-modules $U\colon \mathsf{Mod}_R \to \mathsf{Set}$ is represented by R. Indeed, $\hom(R, M) \cong UM$ by mapping $\phi \mapsto \phi(1)$.

(3) The functor $O_n\colon \mathsf{Grp} \to \mathsf{Set}$ mapping a group to the set of elements of order $\leq n$ is represented by $\mathbb{Z}/n\mathbb{Z}$. A morphism $\hom(\mathbb{Z}/n\mathbb{Z}, G)$ maps to its image on 1 again.

(4) The contravariant powerset functor $\mathcal{P}\colon \mathsf{Set}^{\mathsf{op}} \to \mathsf{Set}$, which maps sets to their powersets and $f\colon X \to Y$ to $f^{-1}\colon \mathcal{P}(Y) \to \mathcal{P}(X)$ is represented by $\{0, 1\}$.

(5) The open-set functor $\mathrm{Open}\colon \mathsf{Top}^{\mathsf{op}} \to \mathsf{Set}$ mapping a topology to its collection of open sets (and $f\colon X \to Y$ maps to $f^{-1}\colon \mathrm{Open}(Y) \to \mathrm{Open}(X)$) is represented by $\{0, 1\}$ with the Sierpenski topology $\{\emptyset, \mathsf{Set}0, \{0, 1\}\}$.

Representations are useful due to the following lemma.

3.2 Yoneda's lemma

For two functors F, G we denote by $\mathrm{Nat}(F, G)$ the set of natural transformations $F \Rightarrow G$ (i.e. Nat is the hom-set of the functor category).

3.2.1 Theorem (Yoneda's Lemma)

Let $F\colon \mathcal{C} \to \mathsf{Set}$ be a functor and $X \in \mathcal{C}$. Then there is a bijection

$$\Phi\colon \mathrm{Nat}(\hom(X, \bullet), F) \xrightarrow{\sim} F(X)$$

which is natural in both X and F.

Proof

We build Φ explicitly: given a natural transformation $\alpha\colon \hom(X, \bullet) \Rightarrow F$, map it to $\alpha_X(1_X)$. To show its bijectivity, we explicitly construct its inverse as well. For $a \in F(X)$ let us define the natural transformation $\alpha\colon \hom(X, \bullet) \Rightarrow F$ whose components are

$$\alpha_Y\colon \hom(X, Y) \to F(Y), \qquad \alpha_Y(f) = F(f)(a)$$

Explicitly, we have

$$\Phi(\alpha) = \alpha_X(1_X), \qquad \Psi(a)_Y(f) = F(f)(a)$$

These are inverses:

$$\Phi(\Psi(a)) = \Psi(a)_X(1_X) = F(1_X)(a) = a, \qquad \Psi(\Phi(\alpha))_Y(f) = F(f)(\Phi(\alpha)) = F(f)\alpha_X(1_X) = \alpha_Y(f)$$

where the final equality is due to naturality: $F(f)\alpha_X = \alpha_Y f_*$ so taking the value of this at 1_X gives $\alpha_Y(f)$.

We show that Φ is natural in X and F separately; first we set F constant and consider Φ_X. To show that Φ_F is natural, let us consider what exactly our functors are. On the right clearly it is F, but on the left it is not as clear. Explicitly, we have a functor $\mathcal{C} \to [\mathcal{C}, \mathsf{Set}]^{\mathsf{op}}$ which maps $X \in \mathcal{C}$ to $h^X = \hom(X, \bullet)$ and f to $f_*\colon \hom(Y, \bullet) \to \hom(X, \bullet), h \mapsto hf$ Since Nat is the hom-functor $\mathrm{Nat}(\bullet, F)\colon [\mathcal{C}, \mathsf{Set}]^{\mathsf{op}} \to \mathsf{Set}$, we see that $\mathrm{Nat}(h^\bullet, F)$ is the composition:

$$\mathcal{C} \to [\mathcal{C}, \mathsf{Set}]^{\mathsf{op}} \xrightarrow{\mathrm{Nat}(\bullet, F)} \mathsf{Set}$$

This maps $f\colon X \to Y$ to f^{**} which maps α to $\alpha \circ f^*$ (whose Z component maps $g\colon Y \to Z$ to $\alpha_Z(gf)$).

So we want to show that $F(f) \circ \Phi_X = \Phi_Y \circ f^{**}$. For $\alpha\colon \hom(X, \bullet) \Rightarrow F$ the left maps to $F(f)(\alpha_X(1_X))$, and the right maps to $(\alpha f^*)_Y(1_Y) = \alpha_Y(f)$. We already showed (when showing $\Psi\Phi(\alpha) = \alpha$) that $F(f)\alpha_X(1_X) = \alpha_Y(f)$, as desired.

Now we set X and we consider Φ_F as F varies. The left-hand side is simply the covariant hom-functor represened by $\hom(X, \bullet)$, while the right-hand side is the functor $\mathrm{ev}_X\colon [\mathcal{C}, \mathsf{Set}] \to \mathsf{Set}$ which evaluates $F\colon \mathcal{C} \to \mathsf{Set}$ at $F(X)$, and $\lambda\colon F \Rightarrow G$ is mapped to $\mathrm{ev}_X(\lambda) = \lambda_X\colon F(X) \to G(X)$. Thus naturality just states that $\lambda_X \circ \Phi_F = \Phi_G \circ \lambda_*$ for all $\lambda\colon F \Rightarrow G$. Indeed, if $\alpha\colon \hom(X, \bullet) \Rightarrow F$ is natural its image on the left is $\lambda_X\alpha_X(1_X)$ and its image on the right is $\Phi_G(\lambda \circ \alpha) = (\lambda_X\alpha_X)(1_X)$, which is equal. ■

Yoneda's lemma dualizes to:

3.2.2 Corollary

Let $F\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ be a contravariant functor and $X \in \mathcal{C}$, then there is a natural bijection

$$\mathrm{Nat}(\hom(\bullet, X), F) \cong FX, \qquad \alpha \mapsto \alpha_X(1_X)$$

In our proof we defined a functor $\mathcal{C} \to [\mathcal{C}, \mathsf{Set}]^{\mathsf{op}}$ which mapped $X \in \mathcal{C}$ to $\hom(X, \bullet)$ and $f\colon X \to Y$ to $f_*\colon \hom(Y, \bullet) \Rightarrow \hom(X, \bullet)$. A functor $\mathcal{C} \to \mathcal{D}^{\mathsf{op}}$ is equivalent to a functor $\mathcal{C}^{\mathsf{op}} \to \mathcal{D}$ (with the same definition),

so this is simply a functor $\mathcal{C}^{\mathsf{op}} \to [\mathcal{C}, \mathsf{Set}]$. Replacing $\mathcal{C}$ with its opposite category, we get a functor $\mathcal{C} \to [\mathcal{C}^{\mathsf{op}}, \mathsf{Set}]$ defined as follows:

(1) $X \in \mathcal{C}$ is mapped to $\hom(\bullet, X)$,

(2) $f\colon X \to Y$ is mapped to $f_*\colon \hom(\bullet, X) \Rightarrow \hom(\bullet, Y)$ (composition with f).

3.2.3 Definition

The functor $\mathcal{C} \to [\mathcal{C}^{\mathsf{op}}, \mathsf{Set}]$ just described is called the **Yoneda embedding**, and it is denoted y. We denote the related functor $\mathcal{C}^{\mathsf{op}} \to [\mathcal{C}, \mathsf{Set}]$ (the Yoneda embedding for the opposite category) by y^{op} (or just y when no confusion arises).

3.2.4 Proposition

y is a fully faithful embedding.

Proof

To show that y is fully faithful, we must consider for $X, Y \in \mathcal{C}$ the enduced function $\mathsf{y}_{XY}\colon \hom(X, Y) \to \mathrm{Nat}(\mathsf{y}(X), \mathsf{y}(Y))$ and show it is a bijection. Notice that $\mathrm{Nat}(\mathsf{y}(X), \mathsf{y}(Y)) \cong \hom(X, Y)$ by the Yoneda lemma, which maps $\alpha\colon \mathsf{y}(X) \Rightarrow \mathsf{y}(Y)$ to $\alpha_X(1_X)$. We know that this is a bijection, so it is sufficient to show that composition with y_{XY} is the identity (in either direction) in order to show that y_{XY} is a bijection. Let $f\colon X \to Y$, then $\mathsf{y}(f)_X(1_X) = f$ by definition, as desired. ■

Since fully faithful functors reflect isomorphisms:

3.2.5 Proposition

For any two objects $X, Y \in \mathcal{C}$, $\hom(X, \bullet) \cong \hom(Y, \bullet)$ iff $X \cong Y$. Similarly for represented functors. Thus, a representation of a functor is unique up to isomorphism.

3.2.6 Definition

A contravariant functor $\mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ is called a **presheaf**. We denote the category of presheaves (i.e. $[\mathcal{C}^{\mathsf{op}}, \mathsf{Set}]$) by $\mathbf{Psh}(\mathcal{C})$.

So the Yoneda embedding gives a fully faithful embedding $\mathcal{C} \to \mathbf{Psh}(\mathcal{C})$.
Recall that the center of a category is the commutative monoid of natural transformations $1_C \Rightarrow 1_C$. Further recall that the center of equivalent categories are isomorphic.

3.2.7 Example

We claim that Grp and Ab are not equivalent, we prove this by computing their centers. Recall that the forgetful functor $U\colon \mathsf{Grp} \to \mathsf{Ab}$ is represented by $\mathbb{Z}$, and it is faithful. As it is faithful, it induces and embedding $Z(\mathsf{Grp}) \to \mathrm{Nat}(U, U) \cong \hom(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}$. The bijection is given by $n \in \mathbb{Z}$ maps to the natural transformations whose components are given by $x \mapsto x^n$. The only one of these natural transformations whose components are group morphisms are when $n = 0, 1$. Thus $Z(\mathsf{Grp}) = \{0, 1\}$ with multiplication.

This shows that $Z(\mathsf{Ab}) = \mathbb{Z}$ since $x \mapsto x^n$ are all group morphisms. Thus $Z(\mathsf{Ab}) \cong Z(\mathsf{Grp})$ and so they cannot be equivalent.

3.3 Representables and (co)limits

3.3.1 Proposition

Let $F\colon \mathcal{C} \to \mathsf{Set}$, then F is corepresentable iff $\int F$ has an initial object. Dually $F\colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ is representable iff $\int F$ has a terminal object.

Proof

Suppose that $(c, x) \in \int F$ is initial. Recall the proof of the Yoneda lemma, in which we define the natural transformation $\Psi(x)\colon \hom(c, \bullet) \Rightarrow F(c)$ by $\Psi(x)_d(f) = Ff(x)$. For each $d \in \mathcal{C}$ and $y \in F(d)$ there is a unique $f\colon c \to d$ such that $Ff(x) = y$. Thus this f is the unique $f\colon c \to d$ such that $\Psi(x)_d(f) = y$. Thus $\Psi(x)_d$ is a bijection for each $d \in \mathcal{C}$, and thus $\Psi(x)$ is a natural isomorphism represented F.

Now suppose $\alpha\colon \hom(c, \bullet) \Rightarrow F$ is an isomorphism. Consider then $(c, \Phi(\alpha) = \alpha_c(1_c)) \in \int F$: it is initial. Indeed, let $(d, x) \in \int F$; a morphism $(c, \Phi(\alpha)) \to (d, x)$ is a morphism $f\colon c \to d$ such that $x = Ff(\Phi(\alpha)) = \Psi(\Phi(\alpha))_d(f) = \alpha_d(f)$. Such a morphism exists and is unique since $\alpha_d\colon \hom(c, d) \to Fd$ is a bijection. Thus $(c, \Phi(\alpha))$ is initial. ■

Since limits are terminal objects in the category of elements of $\mathrm{Cone}(\bullet, F)$, we get the following:

3.3.2 Corollary

$F\colon I \to \mathcal{C}$ has a limit $\lim_I F$ if and only if $\lim_I F$ represents $\mathrm{Cone}(\bullet, F)$ (i.e. $\hom(\bullet, \lim_I F) \cong \mathrm{Cone}(\bullet, F)$). Dually $F\colon I \to \mathcal{C}$ has a colimit $\mathrm{colim}_I F$ iff $\mathrm{colim}_I F$ corepresents $\mathrm{Cone}(F, \bullet)$.

Given a diagram $F\colon I \to \mathcal{C}$ and an element $X \in \mathcal{C}$, consider the functor $\hom(X, F\bullet)\colon I \to \mathsf{Set}$. Since Set is complete, this has a limit $\lim_I \hom(X, F\bullet)$. Recall that the limit of a diagram $F\colon I \to \mathsf{Set}$ is the equalizer of

$$f, g\colon \prod_{i \in I} F(i) \rightrightarrows \prod_{\alpha : i \to j} F(j), \qquad f = (p_j)_{\alpha : i \to j},\ g = (F(\alpha) p_i)_{\alpha : i \to j}$$

which is simply

$$\lim_I G = \{(a_i)_{i \in I} \mid \forall \alpha\colon i \to j.\, a_j = F(\alpha)(a_i)\}$$

So in our case

$$\lim_I \hom(X, F\bullet) = \{(\lambda_i\colon X \to F(i))_{i \in I} \mid \lambda_j = F(\alpha) \circ \lambda_i\}$$

this is precisely the set of cones $X \Rightarrow F$, i.e. $\lim_I \hom(X, F\bullet) \cong \mathrm{Cone}(X, F)$. This isomorphism is natural. Thus if F has a limit $\lim_I F$ then it represents $\mathrm{Cone}(\bullet, F)$, i.e.

$$\lim_I \hom(X, F\bullet) \cong \mathrm{Cone}(X, F) \cong \hom(X, \lim_I F)$$

naturally. This isomorphism also preserves the limit cone, so we have that $\hom(X, \bullet)$ preserves limits. Dually, $\hom(\bullet, X)$ preserves limits in the opposite category, i.e. it carries colimits to limits.

3.3.3 Proposition

The covariant hom-functors $\hom(X, \bullet)$ preserve limits, while contravariant hom-functors $\hom(\bullet, X)$ carry colimits to limits.

Since isomorphic functors preserve the same limits, we get

3.3.4 Proposition

Representable and corepresentable functors preserve limits (representable functors carry colimits to limits,

as their codomain is the opposite category).

4 Adjunctions

Consider the concept of a "free" functor. For example, the functor $\mathbb{Z}[\bullet]\colon \mathsf{Set} \to \mathsf{Ab}$ which maps a set X to the free Abelian group on X; $\mathbb{Z}[X]$ is simply the set of formal integer combinations of elements from X. This is free in the sense that every morphism $\mathbb{Z}[X] \to A$ factors uniquely through a pure set-theoretic function $X \to A$, and vice versa (every such function gives rise to such a morphism). In particular, this gives us a natural isomorphism

$$\hom_{\mathsf{Ab}}(\mathbb{Z}[X], A) \cong \hom_{\mathsf{Set}}(X, UA)$$

where U is the forgetful functor. As it turns out such an occurrence is quite important in category theory and beyond.

4.0.1 Definition

An **adjunction** is a pair of functors $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon G$ such that there exists a natural isomorphism

$$\Phi_{cd}\colon \hom_{\mathcal{D}}(Fc, d) \xrightarrow{\sim} \hom_{\mathcal{C}}(c, Gd)$$

considering both sides as functors $\mathcal{C}^{\mathsf{op}} \times \mathcal{D} \to \mathsf{Set}$. In such a case F is called the **left adjoint** and G the **right adjoint**. This relation is denoted $F \dashv G$.

Note that $F \dashv G$ forms an adjunction iff $G^{\mathsf{op}} \dashv F^{\mathsf{op}}$ does, so there is a duality between left and right adjoints.

Recall that given a bifunctor $F\colon \mathcal{C} \times \mathcal{D} \to \mathcal{E}$, for every $c \in \mathcal{C}, d \in \mathcal{D}$ we have functors $F(c, \bullet)\colon \mathcal{C} \to \mathcal{E}$ and $F(\bullet, d)\colon \mathcal{D} \to \mathcal{E}$. Furthermore, if $f\colon c \to c'$ then we have natural transformations $F(f, \bullet)\colon F(c, \bullet) \Rightarrow F(c', \bullet)$ (similar for $\mathcal{D}$) defined by $F(f, \bullet)_d = F(f, 1_d)$. This is simply currying: $[\mathcal{C} \times \mathcal{D}, E] \cong [\mathcal{C}, [\mathcal{D}, \mathcal{E}]]$.

In particular for the bifunctor $\hom_C\colon \mathcal{C}^{\mathsf{op}} \times \mathcal{C} \to \mathsf{Set}$ we have the usual hom-functors. For $f\colon c \to c'$ these natural transformations are $f^* = \hom(f, \bullet)\colon \hom(c', \bullet) \Rightarrow \hom(c, \bullet)$ $(g \mapsto gf)$ and $f_* = \hom(\bullet, f)\colon \hom(\bullet, c) \Rightarrow \hom(\bullet, c')$ $(g \mapsto fg)$. Note that these are simply the images of f under the respective Yoneda embeddings.

4.0.2 Theorem

Right (and thus by duality left) adjoints are unique up to unique isomorphism. Explicitly, if $F\colon \mathcal{C} \to \mathcal{D}$ has two right adjoints G, G' (with isomorphisms Φ, Φ') then there exists a unique isomorphism $\theta\colon G \to G'$ such that for every d, $(\theta_d)_* \circ \Phi_d = \Phi'_d$ (so for every $f\colon Fc \to d$, $\Phi'(f) = \theta_d \circ \Phi(f)$).

Proof

We require that $(\theta_d)_* = \Phi'_d \circ \Phi_d^{-1}$, and since the Yoneda embedding is fully faithful there exists a unique θ_d for each such d which satisfy this (i.e. $f_* = g_*$ iff $f = g$), given by $\theta_d = \Phi'_{Gd,d} \circ \Phi^{-1}_{Gd,d}(1_{Gd})$. θ_d is an isomorphism (since fully faithful functors reflect isomorphisms, and $\Phi'_d \circ \Phi^1_d$ is clearly an isomorphism). All that remains to show is that θ forms a natural transformation.

For $f\colon d \to d'$ we need to show that $G'f \circ \theta_d = \theta_{d'} \circ Gf$. Since the Yoneda embedding is fully faithful, this holds iff the equation holds when taking the Yoneda embedding of it: $(G'f)_* \circ \Phi'_d \circ \Phi_d^{-1} = \Phi'_{d'} \circ \Phi^1_{d'} \circ (Gf)_*$. This holds by Φ, Φ''s naturality. ■

4.0.3 Example

An extremely common example of an adjunction is a "free $\dashv$ forgetful" adjunction. The example we gave at the beginning is one such adjunction, others include:

(1) $k[\bullet]\colon \mathsf{Set} \to \mathsf{Vec}_k$ which maps a set X to the free vector space on X: all k-linear combinations of X.

(2) $\mathbb{Z}[\bullet]\colon \mathsf{Set} \to \mathsf{CRing}$ which maps a set X to the set of all polynomials with coefficients in $\mathbb{Z}$ whose variables commute.

(3) $F[\bullet]\colon \mathsf{Set} \to \mathsf{Grp}$ which maps a set to the free group on that set.

(4) $(\bullet)_*\colon \mathsf{Set} \to \mathsf{Set}_*$ which maps a set X to the pointed set $(X \sqcup \mathsf{Set}*, *)$.

4.0.4 Example

Some other types of adjunctions are:

(1) The forgetful functor $U\colon \mathsf{Top} \to \mathsf{Set}$ has both a left and right adjoint. The left adjoint as always is a free functor: a set X maps to the discrete topology on X (every set is open). The right adjoint maps a set X to the trivial space on X (only X and the empty set are open).

(2) For a subgroup $H \subseteq G$ the forgetful functor from G-representations to H-representations $\mathrm{Res}_H^G\colon \mathsf{Vec}_G \to \mathsf{Vec}_H$ also admits left and right adjoints: $\mathrm{ind}_H^G \dashv \mathrm{Res}_H^G \dashv \mathrm{Ind}_H^G$. When G, H are finite then the left and right adjoints are isomorphic. This is an important result in representation theory.

Recall that by currying we have a natural isomorphism:

$$\hom_{\mathsf{Set}}(X \times Y, Z) \cong \hom_{\mathsf{Set}}(X, \hom_{\mathsf{Set}}(Y, Z))$$

which is natural in all variables. If we set Y constant, this gives us an adjunction

$$\cdot \times Y \dashv \hom(Y, \bullet)$$

Similarly for Abelian groups, the tensor product gives us a similar adjunction:

$$\hom_{\mathsf{Ab}}(A \otimes B, C) \cong \hom_{\mathsf{Ab}}(A, \hom_{\mathsf{Ab}}(B, C))$$

so we have

$$\cdot \otimes B \dashv \hom(B, \bullet)$$

4.1 The unit and counit

Consider a free functor $F\colon \mathcal{C} \to \mathcal{D}$: this means that for every $X \in \mathcal{C}$ there is a map $\eta_X\colon X \to UFX$ such that every map $FX \to A$ factors through it. That is, for every $f\colon FX \to A$ there is a unique $f'\colon X \to UA$ such that $f' = U(f) \circ \eta_X$.

What is this η_X? Well it is natural $\eta\colon 1_{\mathcal{C}} \Rightarrow UF$, and it defines the adjunction's isomorphism: $\hom(FX, A) \to \hom(X, UA)$ by $f \mapsto U(f) \circ \eta_X$. So it is natural to ask, is this a one-time occurrence? As it turns out, no; it is an equivalent way of characterizing adjunctions.

4.1.1 Lemma

Let $\Phi_{cd}\colon \hom(Fc, d) \to \hom(c, Gd)$, then Φ's naturality is equivalent to the property that the following left diagram commutes iff the right one does (for every choice of objects and morphisms):

$$\begin{tikzcd} F(X_1) \arrow[r, "\alpha"] \arrow[d, "F(f)"'] & Y_1 \arrow[d, "g"] \\ F(X_2) \arrow[r, "\beta"] & Y_2 \end{tikzcd} \qquad \begin{tikzcd} X_1 \arrow[r, "\Phi(\alpha)"] \arrow[d, "f"] & G(Y_1) \arrow[d, "G(g)"'] \\ X_2 \arrow[r, "\Phi(\beta)"] & G(Y_2) \end{tikzcd}$$

Proof

For $f = 1_X$ we get that the condition means:

$$\beta = g\alpha \iff G(g)\Phi(\alpha) = \Phi(\beta)$$

i.e.

$$G(g)\Phi(\alpha) = \Phi(g\alpha)$$

Which is just equivalent to Φ's naturality in its second component. Similar for $g = 1_Y$. Thus this condition implies Φ's naturality, and the other direction is a usual computation. ■

For an adjunction $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon G$ we have a natural isomorphism

$$\Phi_{X,FX}\colon \hom(FX, FX) \cong \hom(X, GFX)$$

and so we get a morphism $\eta_X = \Phi_{X,FX}(1_{FX})\colon X \to GFX$.

4.1.2 Lemma

η_X forms a natural transformation $1_{\mathcal{C}} \Rightarrow GF$.

Proof

We need to show that for every $f\colon X \to Y$, $\eta_Y f = GF(f)\eta_X$. That is, $\Phi_{Y,FY}(1_{FY})f = GF(f)\Phi_{X,FX}(1_{FX})$, which by the above lemma commutes iff $1_{FY}Ff = Ff1_{FX}$ which obviously does. ■

Similarly we have

$$\Phi_{GY,Y}\colon \hom(FGY, Y) \cong \hom(GY, GY)$$

and so we get $\epsilon_Y = \Phi^{-1}_{GY,Y}(1_{GY})\colon FGY \to Y$, which is also natural $FG \Rightarrow 1_{\mathcal{D}}$.

4.1.3 Definition

Let $F \dashv G$ be an adjunction, with Φ the isomorphism. Then the **unit** of the adjunction is the natural transformation $\eta\colon 1_{\mathcal{C}} \Rightarrow GF$ defined by $\eta_X = \Phi_{X,FX}(1_{FX})$. And the **counit** of the adjunction is the natural transformation $\epsilon\colon FG \Rightarrow 1_{\mathcal{D}}$ by $\epsilon_Y = \Phi^{-1}_{GY,Y}(1_{GY})$.

4.1.4 Lemma (The triangle identities)

For an adjunction $F \dashv G$, its unit-counit pair η, ϵ satisfies the following **triangle identites**: the compositions

$$F \xrightarrow{F\eta} FGF \xrightarrow{\epsilon F} F, \qquad G \xrightarrow{\eta G} GFG \xrightarrow{G\epsilon} G$$

are the identities.

Proof

We want to show hat $\epsilon_{FX} \circ F(\eta_X) = 1_{FX}$ for every $X \in \mathcal{C}$. Note that this is equivalent to $\epsilon_{FX} \circ F(\eta_X) = F(1_X) \circ 1_{FX}$, i.e. $\Phi^{-1}(1_{GFX}) \circ F(\eta_X) = F(1_X) \circ \Phi^{-1}(\eta_X)$. By the above lemma this is just equivalent to $1_{GFX} \circ \eta_X = 1_X \circ \eta_X$ which is clearly true. The second identity follows by symmetry. ■

4.1.5 Theorem

Let $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon G$ be a pair of functors, with natural transformations $\eta\colon 1_{\mathcal{C}} \Rightarrow GF$ and $\epsilon\colon FG \Rightarrow 1_{\mathcal{D}}$ which satisfy the above triangle identities. Then $F \dashv G$ forms an adjunction (with unit-counit pair η, ϵ).

Proof

We define $\Phi\colon \hom(FX, Y) \to \hom(X, GY)$ by the composition $\Phi(f)\colon X \xrightarrow{\eta_X} GF(X) \xrightarrow{G(f)} GY$. Similarly we define its inverse $\Psi\colon \hom(X, GY) \to \hom(FX, Y)$ by $\Psi(g)\colon FX \xrightarrow{Fg} FG(Y) \xrightarrow{\epsilon_Y} Y$. These are indeed inverses:

$$\Psi(\Phi(f)) = \Psi(G(f)\eta_X) = \epsilon_Y FG(f)F(\eta_X)$$

Now since $\epsilon_Y \circ FG(f) = f \circ \epsilon_{FX}$ by naturality, we get that this is equal to $f\epsilon_{FX}F(\eta_X) = f$ by the triangle identity. Similarly $\Phi(\Phi(g)) = g$, so they are indeed inverses.

By definition $\Phi(1_{FX}) = \eta_X$ and $\Psi(1_{GY}) = \epsilon_Y$, so this isomorphism gives η, ϵ as the unit-counit pair. ■

4.1.6 Proposition

If η, ϵ satisfy the triangle identities, then η only satisfies them with ϵ, and ϵ only with η.

Proof

Notice that above we defined Φ only using η. Thus if ϵ' also satisfies the triangle identities, then $\Phi^{-1}(g)$ is given by both $\epsilon_Y F(g)$ and $\epsilon'_Y F(g)$. In particular they are equal for $g = 1$, and so $\epsilon_Y = \epsilon'_Y$. ■

Notice that if $f\colon X \to Y$, then

$$\Phi(f \circ \epsilon_X) = G(f)G(\epsilon_X)\eta_{GX} = G(f)$$

so we can recover G's action on morphisms by the unit and counit.

4.1.7 Lemma

The following are all equivalent:

(1) $F \dashv G$,

(2) There is a natural isomorphism $\hom_{\mathcal{D}}(FX, Y) \cong \hom_{\mathcal{C}}(X, GY)$,

(3) There are natural transformations $\eta\colon 1_{\mathcal{C}} \Rightarrow GF$ and $\epsilon\colon FG \Rightarrow 1_{\mathcal{D}}$ satisfying the triangle identities,

(4) There is a natural transformation $\eta\colon 1_{\mathcal{C}} \Rightarrow GF$ such that

$$\hom_{\mathcal{D}}(FX, Y) \xrightarrow{G} \hom_{\mathcal{C}}(GFX, GY) \xrightarrow{\circ\eta_X} \hom_{\mathcal{C}}(X, GY)$$

forms an isomorphism,

(5) There is a natural transformation $\epsilon\colon FG \Rightarrow 1_{\mathcal{D}}$ such that

$$\hom_{\mathcal{C}}(X, GY) \xrightarrow{F} \hom_{\mathcal{D}}(FX, FGY) \xrightarrow{\epsilon_Y \circ} \hom_{\mathcal{D}}(FX, Y)$$

forms an isomorphism.

Proof

(1) and (2) are equivalent by definition, and we showed they are equivalent to (3). Furthermore we showed that they imply (5), and by symmetry (4), and these both clearly imply (2). ■

4.1.8 Proposition

Consider four functors $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D} \underset{G'}{\overset{F'}{\rightleftarrows}} \mathcal{E}$ which form adjunctions $F \dashv G$, $F' \dashv G'$. Then $F'F \dashv GG'$ forms an adjunction $\mathcal{C} \rightleftarrows \mathcal{E}$.

Proof

The composition of isomorphisms

$$\hom_{\mathcal{E}}(F'Fc, e) \cong \hom_{\mathcal{D}}(Fc, G'e) \cong \hom_{\mathcal{C}}(c, GG'e)$$

is a natural isomorphism. If the isomorphisms are given by Φ, Φ', this new one is given by

$$\hat{\Phi}_{c,e} = \Phi_{c,G'e} \circ \Phi'_{Fc,e}$$

■

We can then ask ourselves, what is the unit-counit pair of the composition of adjunctions? Suppose $(\eta, \epsilon)\colon F \dashv G$ and $(\eta', \epsilon')\colon F' \dashv G'$ form adjunctions, then they form isomorphisms $\Phi_{c,d}(f) = G(f)\eta_c$ and $\Phi'_{d,e}(f) = G'(f)\eta'_d$ with inverses $\Psi_{c,d}(g) = \epsilon_d F(g)$ and $\Psi'_{d,e}(g) = \epsilon'_e F'(g)$. Then if $\hat{\Phi}$ and $\hat{\Psi}$ are the isomorphisms of the composition and $\hat{\eta}, \hat{\epsilon}$ the unit-counit pair, we have:

$$\hat{\eta}_c = \hat{\Phi}_{c,F'Fc}(1_{F'Fc}) = \Phi_{c,G'F'Fc}\Phi'_{Fc,F'Fc}(1_{F'Fc}) = \Phi_{c,G'F'Fc}(\eta'_{Fc}) = G(\eta'_{Fc})\eta_c$$

and similarly

$$\hat{\epsilon}_e = \epsilon'_e F'(\epsilon_{G'e})$$

Using whiskering we have the following:

4.1.9 Proposition

Given adjunctions $(\eta, \epsilon)\colon F \dashv G$ and $(\eta', \epsilon')\colon F' \dashv G'$, we have $(G\eta' F \circ \eta, \epsilon' \circ F'\epsilon G')\colon F'F \dashv GG'$.

4.2 Equivalences and adjunctions

4.2.1 Proposition

Let $F \dashv G$ be an adjunction.

(1) G is fully faithful iff the counit $\epsilon\colon FG \Rightarrow 1_{\mathcal{D}}$ is an isomorphism.

(2) F is fully faithful iff the unit $\eta\colon 1_{\mathcal{C}} \Rightarrow GF$ is an isomorphism.

Proof

Recall that G's action on morphisms is given by

$$\hom_{\mathcal{C}}(X, Y) \xrightarrow{\circ\epsilon_X} \hom_{\mathcal{C}}(FGX, Y) \xrightarrow{\Phi} \hom_{\mathcal{D}}(GX, GY)$$

Since Φ is always a bijection, this is a bijection iff $\circ\epsilon_X$ is a bijection. This is just the image of ϵ_X under the Yoneda embedding, and as such it is a bijection iff ϵ_X is a bijection. Thus G is fully faithful iff ϵ is an isomorphism. ■

4.2.2 Definition

An adjunction $F \dashv G$ is an **adjoint equivalence** if F and G are equivalences.

Note that we are implicitly including a unit-counit pair or an isomorphism Φ in our definition here. Recall that an equivalence comes with isomorphisms $\eta\colon 1_{\mathcal{C}} \Rightarrow GF$ and $\epsilon\colon FG \Rightarrow 1_{\mathcal{D}}$ by definition. But these don't necessarily satisfy the triangle identities!

4.2.3 Proposition

Every equivalence (F, G, η, ϵ) can be upgraded to an adjoint equivalence $F \dashv G$ such that the unit (or counit) stays the same.

Proof

We can form an isomorphism:

$$\hom_{\mathcal{D}}(Fc, d) \xrightarrow{G} \hom_{\mathcal{C}}(GFc, Gd) \xrightarrow{\circ\eta_c} \hom_{\mathcal{C}}(c, Gd)$$

(this is an isomorphism since η_c is and G is fully faithful.) The unit of this is η, but its counit is not necessarily ϵ. Similarly we can give an isomorphism whose counit is ϵ. ■

4.3 Adjunctions and (co)limits

Recall that a cone $c \Rightarrow K$ can be viewed as a natural transformation $\hom_{\mathcal{C}}(\bullet, c) \Rightarrow \mathrm{Cone}(\bullet, K)$ via the Yoneda lemma. This correspondence is given by

(1) Mapping a cone $\lambda\colon c \Rightarrow K$ to $\Psi(\lambda)_d(f) = \lambda f$ for $d \in \mathcal{C}$ and $f\colon d \to c$.

(2) Mapping a natural transformation $\alpha\colon \hom(\bullet, c) \Rightarrow \mathrm{Cone}(\bullet, K)$ to $\alpha_c(1_c)$.

The limit of K is just a natural isomorphism $\hom(\bullet, c) \cong \mathrm{Cone}(\bullet, K)$.
Following this, if $F\colon \mathcal{C} \to \mathcal{D}$ is a functor then to a cone $\lambda\colon c \Rightarrow K$ we have its image $F\lambda\colon Fc \Rightarrow K$. What is the associated natural transformation? It is $\Psi(F\lambda)\colon \hom_{\mathcal{D}}(\bullet, Fc) \Rightarrow \mathrm{Cone}(\bullet, FK)$ defined by

$$\Psi(F\lambda)_d(f) = F(\lambda)f$$

for all $d \in \mathcal{D}$ and $f: d \to Fc$ So $F\alpha$ is the natural transformation

$$(F\alpha)_d(f) = F(\alpha_c(1_c))f$$

Furthermore recall that $\mathrm{Cone}(\bullet, K)$ is simply the composition

$$\mathrm{Cone}(\bullet, K) = \mathcal{C}^{\mathsf{op}} \xrightarrow{\Delta_I^{\mathsf{op}}} [I, \mathcal{C}]^{\mathsf{op}} \xrightarrow{\mathrm{Nat}(\bullet, K)} \mathsf{Set}$$

where Δ_I is the diagonal functor. So the limit of K is just a representation $\hom(\bullet, c) \cong \mathrm{Nat}(\Delta\bullet, K)$.

Now suppose K has a limit $\lim K$ and $F \lim K$ is a limit for FK. This is not enough to show that F preserves limits: the limit cone needs to be preserved as well. Suppose we have an isomorphism $\alpha: \hom(\bullet, F \lim K) \cong \mathrm{Nat}(\Delta\bullet, FK)$, in order to have F preserve limits, we need α to induce the $F\lambda$ where $\lambda: \lim K \Rightarrow K$ is the limit cone of K. By Yoneda this is iff $\alpha_{F \lim K}(1_{F \lim K}) = F\lambda$.

Further recall that to a morphism $f: c \to d$ we have $f^*: \hom(\bullet, c) \to \hom(\bullet, d)$ natural. In the case that $\mathcal{C}$ is the category of categories, then for a functor $F: \mathcal{C} \to \mathcal{D}$ we get $F^*: [\bullet, \mathcal{C}] \to [\bullet, \mathcal{D}]$. These give us an action on objects $F^I: [I, \mathcal{C}] \to [I, \mathcal{D}]$ ($F^I(K) = FK$). But how should it act on morphisms (natural transformations)? If $\alpha: K \Rightarrow K'$ is a natural transformation then $F^I(\alpha): FK \Rightarrow FK'$ is defined by $F^I(\alpha) = F\alpha$. This defines a functor $F^I: [I, \mathcal{C}] \to [I, \mathcal{D}]$ natural in I.

4.3.1 Proposition

Let $F \dashv G$ form an adjunction and let I be a small category. Then $F^I \dashv G^I$.

Proof

Let $K \in [I, \mathcal{C}]$ and $L \in [I, \mathcal{D}]$ then $\hom(FK, L) \cong \hom(K, GL)$ by mapping a natural transformation $(\alpha_i): FK \Rightarrow L$ to $(\Phi\alpha_i): K \Rightarrow GL$. This is natural in K, L and is an isomorphism. The unit of this adjunction is $\eta_K = (\Phi 1_{FKi}) = (\eta_{Ki})$. And the counit is $\epsilon_L = (\epsilon_{Li})$. ■

We will omit the superscript and just write F for the induced functor.

4.3.2 Theorem

Left adjoints preserve colimits and right adjoints preserve limits.

Proof

Let $F \dashv G$ be an adjunction with unit-counit η, ϵ. Let $K: I \to \mathcal{D}$ be a diagram with a limit: $\lambda: \lim_I K \Rightarrow K$ Let $\alpha: \mathrm{Nat}(\Delta\bullet, K) \cong \hom(\bullet, \lim_I K)$ be the corresponding isomorphism (so $\lambda = \alpha_{\lim K}(1_{\lim K})$, and $\alpha_d(f) = \lambda f$).

Then we have an isomorphism (since $F \dashv G$ in the functor categories):

$$\mathrm{Nat}(\Delta\bullet, GK) \cong \mathrm{Nat}(F\Delta\bullet, K) \cong \mathrm{Nat}(\Delta F\bullet, K) \cong \hom(F\bullet, \lim_I K) \cong \hom(\bullet, G \lim_I K)$$

The first isomorphism is by the adjunction in the functor categories, the second is since $F\Delta = \Delta F$, the third because $\lim_I K$ represents the cone functor, and the final by adjunction (in the normal categories).

This shows that $G \lim_I K \cong \lim_I GK$ (and the right side exists), but it doesn't quite show that G preserves limits: to do that we need to show that $G\lambda$ is a limit cone. To do this we must investigate this isomorphism; in particular we need to watch the inverse image of this isomorphism on $1_{G \lim K}$ and show that it is isomorphic to $G\lambda$. First this maps to $\epsilon_{\lim K}$, then to $(\alpha F)(\epsilon_{\lim K}) = \lambda\epsilon_{\lim K}$, then finally to the adjunct of this: $G(\lambda\epsilon_{\lim K})\eta_{\Delta G \lim K}$. At $i \in I$ this is just $G(\lambda_i\epsilon_{\lim K})\eta_{G \lim K}$, by the triangle identities this is just $G(\lambda_i)$, as desired. ■

4.4 (Co)limits as adjunctions

4.4.1 Theorem

$\mathcal{C}$ has all limits of shape I iff Δ_I has a right adjoint. Similarly $\mathcal{C}$ has all colimits of shape I iff Δ_I has a left adjoint. These adjoints are $\lim_I, \operatorname{colim}_I$ respectively.

Proof

$\Delta_I: \mathcal{C} \rightleftarrows [I, \mathcal{C}]: F$ is an adjunction iff $\operatorname{Nat}(\Delta\bullet, K) \cong \hom(\bullet, F(K))$ is an isomorphism. This means that for every $K: I \to \mathcal{C}$, $\operatorname{Nat}(\Delta\bullet, K) = \operatorname{Cone}(\bullet, K)$ is representable, i.e. K has a limit. And similarly if $\lim_I$ exists then the representation by $\lim_I K$ is natural, so it forms an adjunction. ■

Let us consider $\Delta_{I\times J}: \mathcal{C} \to [I \times J, \mathcal{C}]$, $\Delta_J: \mathcal{C} \to [J, \mathcal{C}]$, and $\Delta_I: [J, \mathcal{C}] \to [I, [J, \mathcal{C}]]$. If $\Phi: [I \times J, \mathcal{C}] \to [I, [J, \mathcal{C}]]$ is currying, then we have $\Phi\Delta_{I\times J} = \Delta_I\Delta_J$. Since adjoints are unique, $\Phi \lim_{I\times J}$ must be right adjoint to $\Delta_I\Delta_J$ which already has right adjoint $\lim_J \lim_I$. So they must be isomorphic, thus

$$\lim_{I\times J} F(i, j) \cong \lim_J \lim_I F(i, j) \cong \lim_I \lim_J F(i, j)$$

4.5 Per-object adjunctions

Consider an adjunction $F: \mathcal{C} \rightleftarrows \mathcal{D}: G$, giving a natural isomorphism $\Phi: \hom_{\mathcal{D}}(Fc, d) \cong \hom_{\mathcal{C}}(c, Gd)$. If we set $d \in \mathcal{D}$ constant, this means that $\hom_{\mathcal{C}}(\bullet, Gd)$ is a representation of the composite functor $\hom_{\mathcal{D}}(F\bullet, d)$. This gives rise to the following definition:

4.5.1 Definition

A functor $F: \mathcal{C} \to \mathcal{D}$ has a **right adjoint at** $d \in \mathcal{D}$ if the functor $\hom_{\mathcal{D}}(F\bullet, d): \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ is representable.

4.5.2 Theorem

A functor $F: \mathcal{C} \to \mathcal{D}$ has a right adjoint iff it has a right adjoint at every object in $\mathcal{D}$. In particular, if for every $d \in \mathcal{D}$ we choose a representation $G(d) \in \mathcal{C}$ for $\hom_{\mathcal{D}}(F\bullet, d)$, then we can extend G to a right adjoint functor of F.

Proof

If F has a right adjoint then this is clear. Conversely, consider the Yoneda embedding $\mathsf{y}_{\mathcal{D}}: \mathcal{D} \to [\mathcal{D}^{\mathsf{op}}, \mathsf{Set}]$, and compose it with $\circ F^{\mathsf{op}}: [\mathcal{D}^{\mathsf{op}}, \mathsf{Set}] \to [\mathcal{C}^{\mathsf{op}}, \mathsf{Set}]$. That is, define $\hat{G}: \mathcal{D} \to [\mathcal{C}^{\mathsf{op}}, \mathsf{Set}]$ by $\hat{G} = (\circ F^{\mathsf{op}})\mathsf{y}_{\mathcal{D}}$. So $\hat{G}$ satisfies

$$\hat{G}(d)(\bullet) = \hom(F\bullet, d)$$

By assumption for every $d \in \mathcal{D}$, $\hat{G}(d) \cong \hom(\bullet, G(d)) = \mathsf{y}_{\mathcal{C}}(G(d))$, so $\hat{G}$'s image lies in $\mathsf{y}_{\mathcal{C}}$'s image (up to isomorphism). Since $\mathsf{y}_{\mathcal{C}}$ is fully faithful and thus an equivalence on its image, we get a functor $G: \mathcal{D} \to \mathcal{C}$ such that $\mathsf{y}_{\mathcal{C}} \circ G \cong \hat{G}$. ■

Now suppose we want to represent $\hom_{\mathcal{D}}(F\bullet, d)$. So we are looking for $G(d)$ and a natural isomorphism $\Psi_d: \hom(\bullet, Gd) \to \hom(F\bullet, d)$. By Yoneda, natural transformations α of this sort are in one-to-one correspondence with elements of $\hom(FGd, d)$. The correspondence given by mapping $\alpha \mapsto \alpha_{Gd}(1_{Gd})$, denote it by $\hat{\alpha}$.

Recall that we can write α via $\hat{\alpha}$:

$$\alpha_c(f) = \hom(Ff, d)(\hat{\alpha}) = \hat{\alpha} \circ Ff$$

So we can write α as the composition

$$\alpha_c = \hom(c, Gd) \xrightarrow{F} \hom(Fc, FGd) \xrightarrow{\hat{\alpha}\circ} \hom(Fc, d)$$

In particular α is an isomorphism iff this is an isomorphism for every $c \in \mathcal{C}$.

4.5.3 Definition

Let $d \in \mathcal{D}$, then $\epsilon_d \colon FGd \to d$ is the **counit at** d if the composition $\epsilon_d \circ F\bullet \colon \hom(\bullet, Gd) \to \hom(F\bullet, d)$ is an isomorphism.

4.5.4 Theorem

$F \colon \mathcal{C} \to \mathcal{D}$ has a right adjoint iff for every $d \in \mathcal{D}$ there exists an object $G(d) \in \mathcal{C}$ with a counit at d $\epsilon_d \colon FGd \to d$. In such a case G can be extended to a right adjoint and ϵ forms the counit of the adjunction.

Dually $G \colon \mathcal{D} \to \mathcal{C}$ has a unit at $c \in \mathcal{C}$ if there is a map $\eta_c \colon c \to GFc$ such that $G \bullet \circ \eta_c$ is an isomorphism. Then G has a left adjoint iff for every $c \in \mathcal{C}$ there is a unit at c.

4.5.5 Definition

Consider two functors $\mathcal{D} \xrightarrow{F} \mathcal{C} \xleftarrow{G} \mathcal{E}$, we construct their **comma category** $F \downarrow G$ with

(1) Objects are triples $(d \in \mathcal{D}, e \in \mathcal{E}, f \colon Fd \to Ge \in \mathcal{C})$,

(2) Morphisms $(d, e, f) \to (d', e', f')$ are pairs of morphisms $(h \colon d \to d', k \colon e \to e')$ such that $f' \circ Fh = Gk \circ f$.

(Co)cone categories are examples of comma categories: for a diagram $K \colon I \to \mathcal{C}$ consider it as $K \colon 1 \to [I, \mathcal{C}]$ and then consider the comma category

$$\Delta_I \downarrow K : \qquad \mathcal{C} \xrightarrow{\Delta_I} [I, \mathcal{C}] \xleftarrow{K} 1$$

Objects are (removing redundancies) $(c \in \mathcal{C}, \lambda \colon \Delta c \Rightarrow K)$ and morphisms $(c, \lambda) \to (d, \mu)$ are morphisms $f \colon c \to d$ such that $\mu f = \lambda$. This is precisely the category of cones over K.

We can extend this: consider $F \colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ as $F \colon 1 \to [\mathcal{C}^{\mathsf{op}}, \mathsf{Set}]$. Then $\int F$ is isomorphic to $\mathsf{y}_{\mathcal{C}} \downarrow F$. Indeed, objects are $(c \in \mathcal{C}, \alpha \colon \mathsf{y}(c) \Rightarrow F)$, and morphisms $(c, \alpha) \to (d, \beta)$ are morphisms $f \colon c \to d$ such that $\beta \circ \mathsf{y}_{\mathcal{C}}(f) = \alpha$. Recall that natural transformations $\alpha \colon \mathsf{y}(c) \Rightarrow F$ are in bijection with elements of Fc, and the condition on morphisms f just requires $Ff(y) = x$. Thus this comma category is isomorphic to the category of elements:

4.5.6 Proposition

Let $F \colon \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ be a presheaf, then

$$\int F \cong \mathsf{y}_{\mathcal{C}} \downarrow F$$

Our particular case of interest though is comma categories from functors of the form $1 \to \mathcal{C} \xleftarrow{G} \mathcal{D}$. A functor $1 \to \mathcal{C}$ is just a choice of object $c \in \mathcal{C}$, and so we get the comma category denoted $c \downarrow G$. We simplify: the category consists of

(1) objects are pairs $(d \in \mathcal{D}, f \colon c \to Gd)$,

(2) morphisms $(d, f) \to (d', f')$ are $\mathcal{D}$-morphisms $g \colon d \to d'$ such that $G(g) \circ f = f'$.

4.5.7 Lemma

Given $G \colon \mathcal{D} \to \mathcal{C}$, a unit at $c \in \mathcal{C}$ is just an initial object in the comma category $c \downarrow G$.

Proof

A map $\eta_c: c \to GF(c)$ is initial iff for every $f: c \to Gd$ there exists a unique $g: Fc \to d$ such that $f = G(g)\eta_X$. That is, iff

$$\hom(Fc, d) \xrightarrow{G} \hom(GFc, Gd) \xrightarrow{\circ\eta_c} \hom(c, Gd)$$

is a bijection. This is precisely the requirement that η_c be a unit at c. ■

4.5.8 Corollary

G has a left adjoint iff for every $c \in \mathcal{C}$, the comma category $c \downarrow G$ has an initial object.

4.6 The adjoint functor theorem

4.6.1 Lemma

The limit of the diagram $1_{\mathcal{D}}: \mathcal{D} \to \mathcal{D}$ is an initial object in $\mathcal{D}$.

Proof

Suppose $\lambda: d \Rightarrow 1_{\mathcal{D}}$ is a limit cone, so we have morphisms $\lambda_c: d \to c$ for every $c \in \mathcal{C}$. Now $\lambda_d: d \to d$ is a cone morphism from λ to itself: $\lambda_c\lambda_d = \lambda_c$ by naturality. Since λ is terminal, this means that λ_d must be the identity.

Now consider another morphism $f: d \to c$, we want to show $f = \lambda_c$. Since λ is natural, we have that $\lambda_c = f \circ \lambda_d = f \circ \mathrm{id}_d = f$. So $\hom(d, c) = \{\lambda_c\}$, meaning d is initial. ■

4.6.2 Definition

A category $\mathcal{D}$ satisfies the **solution set condition** if there exists a small set I and family of objects in $\mathcal{D}$ $\{d_\alpha\}_{\alpha \in I}$ such that for every $d \in \mathcal{D}$ there exists an $\alpha \in I$ and a morphism $d_\alpha \to d$.

4.6.3 Lemma

Let $\mathcal{D}$ be a complete and locally small category. Then $\mathcal{D}$ satisfies the solution set condition iff $\mathcal{D}$ has an initial object.

Proof

If $\mathcal{D}$ has an initial object it clearly satisfies the solution set condition: just take the solution set to be that initial object. Conversely suppose $\{d_\alpha\}_{\alpha \in I}$ is a solution set, and let $\mathcal{D}_0$ be the full subcategory of $\mathcal{D}$ spanned by these objects. Since $\mathcal{D}$ is locally small and I is small, $\mathcal{D}_0$ is small.

Since $\mathcal{D}$ is complete, the inclusion $\iota: \mathcal{D}_0 \to \mathcal{D}$ has a limit, let it be $\lambda: d_0 \Rightarrow \iota$. We will show that we can lift λ to a limit cone $d_0 \Rightarrow 1_{\mathcal{D}}$, and by the previous lemma this shows that $\mathcal{D}$ has an initial object. For $d \notin \mathcal{D}_0$ there exists $d_\alpha \in \mathcal{D}_0$ and a morphism $f_d: d_\alpha \to d$. We define $\lambda_d: d_0 \to d$ to be the composition $d_0 \xrightarrow{\lambda_{d_\alpha}} d_\alpha \xrightarrow{f_d} d$. This extends λ to a collection of morphisms on all of $\mathcal{D}$, but we must show that it is both natural and limiting.

For a morphism $g: d \to d'$, consider $\lambda_d = f_d \circ \lambda_{d_\alpha}$ and $\lambda_{d'} = f_{d'} \circ \lambda_{d_\beta}$. Now consider the pullout P of $d_\beta \xrightarrow{f_{d'}} d' \xleftarrow{g \circ f_d} d_\alpha$. We then further have $\lambda_P = f_P \circ \lambda_{d_\gamma}$.

So we have a diagram:

$$\begin{tikzcd}
d \arrow[dr, "\lambda_\gamma"] \arrow[drr, "\lambda_\beta"] \arrow[ddr, "\lambda_\alpha"'] & & \\
& d_\gamma \arrow[d] & \\
& P \arrow[r] \arrow[d] & d_\beta \arrow[d] \\
& d_\alpha \arrow[r] & d \arrow[r, "g"] & d'
\end{tikzcd}$$

The arrows between $d_0, d_\alpha, d_\beta, d_\gamma$ commute by naturality of λ. So we can write $\lambda_\alpha, \lambda_\beta$ as factoring through $\lambda_P \colon d_0 \to P$. This λ_P is the unique map $d_0 \to P$ through which they factor. By the universal property of the pullout, this whole diagram must commute then. This gives us $g \circ \lambda_d = \lambda_{d'}$, so λ is natural.

Instead of showing that the lift of λ is universal, note that in our proof of the lemma above, it was sufficient to show that $\lambda_{d_0} = \mathrm{id}_{d_0}$. And indeed this must be the case, since λ_{d_0} is a morphism of λ to itself. ■

This in and of itself is not very interesting, but it does allow us to prove a very powerful theorem.

4.6.4 Lemma

For $G \colon \mathcal{D} \to \mathcal{C}$ and $c \in \mathcal{C}$, the forgetful functor $U \colon c \downarrow G \to \mathcal{D}$ creates all limits which exist in $\mathcal{D}$ and are preserved by G. In particular if G preserves limits and $\mathcal{D}$ is complete, $c \downarrow G$ is complete as well.

Proof

Notice that $c \downarrow G$ is isomorphic to the category of elements of the functor $\hom(c, G\bullet) \colon \mathcal{D} \to \mathsf{Set}$. Its objects are $(d \in \mathcal{D}, f \colon c \to Gd)$ and morphisms $(d, f) \to (d', f')$ are morphisms $h \colon d \to d'$ such that $Gh \circ f = f'$. And the forgetful functor $\int \hom(c, G\bullet) \to \mathcal{D}$ creates all limits in $\mathcal{D}$ preserved by $\hom(c, G\bullet)$. Since hom-functors preserve limits, this preserves all limits preserved by G. ■

4.6.5 Theorem

Let $\mathcal{D}$ be a locally small complete category. Then a functor $G \colon \mathcal{D} \to \mathcal{C}$ has a left adjoint iff

(1) G preserves small limits,

(2) For every $c \in \mathcal{C}$, the comma category $c \downarrow G$ satisfies the solution set condition.

Proof

If G has a left adjoint, we know that it preserves small limits and $c \downarrow G$ has an initial object (the unit at c). Recall that showing G has a left adjoint is equivalent to showing that $c \downarrow G$ has an initial object for every $c \in \mathcal{C}$. So we just need to show that $c \downarrow G$ is complete, but that holds because of our previous lemma. ■

Generally people write out what it means for $c \downarrow G$ to satisfy the solution set condition explicitly: for every $c \in \mathcal{C}$ there exists a set of objects $\{d_i\}_{i \in I}$ such that for every $f \colon c \to Gd$ there exists morphisms $h \colon d_i \to d$ and $\phi_i \colon c \to Gd_i$ such that $f = Gh \circ \phi_i$.

If $\mathcal{C}$ is not locally small we need to add the morphisms ϕ_i to the definition (so that the collection of ϕ_i form a set). If $\mathcal{C}$ is locally small then we do not.

4.6.6 Example

The forgetful functor $U\colon \mathsf{Grp} \to \mathsf{Set}$ has a left adjoint. Grp is complete as we know, and U preserves those limits. So we need to show that U satisfies the solution set condition.

So for every $X \in \mathsf{Set}$ we need to find a collection of morphisms $\{f_i\colon X \to UG_i\}_{i\in I}$ such that every $f\colon X \to UG$ factors through one of the f_is. For every $f\colon X \to UG$ consider the subgroup $G_f \subseteq G$ generated by f's image; this has cardinality bound by $2^{|X|} = \kappa$. And we have that f factors through $X \to U(G_f) \to UG$.

So let $\{G_i\}_i$ be the collection of groups of cardinality $\leq \kappa$ (a representative for each isomorphism class). This satisfies the the solution set condition; as such U has a left adjoint $F\colon \mathsf{Set} \to \mathsf{Grp}$. This functor forms the so-called "free group" over a set X.

4.6.7 Example

Let CBool be the category of complete boolean algebras (boolean algebras where every subset has a supremum). CBool is complete, and the forgetful functor $U\colon \mathsf{CBool} \to \mathsf{Set}$ preserves limits, but has no left adjoint.

This is because for any infinite set X we can construct an arbitrarily large complete boolean algebra generated by X. Thus the solution set condition does not hold.

5 Abelian Categories

5.1 Zero morphisms, zero objects

5.1.1 Definition

A category $\mathcal{C}$ is said to have **zero morphisms** if for every $X, Y \in \mathcal{C}$ there exists a morphism $0_{XY}\colon X \to Y$ such that for all morphisms $f\colon X \to Y, g\colon Y \to Z$,

$$0_{YZ} \circ f = g \circ 0_{XY} = 0_{XZ}$$

Notice that the zero morphism 0_{XY} is unique per X, Y. Generally we omit the subscripts and just write e.g. $0 \circ f = g \circ 0 = 0$.

5.1.2 Definition

A **zero object** in a category is an object which is both terminal and initial.

Clearly if a category has a zero object, all terminal objects are initial and vice versa.

5.1.3 Proposition

A category with a zero object has zero morphisms.

Proof

Let $0 \in \mathcal{C}$ be the zero object, then for every $X, Y \in \mathcal{C}$ there is a unique map $0_{XY}: X \to Y$ which factors uniquely through 0:

$$0_{XY} = X \xrightarrow{!} 0 \xrightarrow{!} Y$$

These are clearly zero morphisms. ■

5.1.4 Example

The category of groups has a zero object: the trivial group.

Consider a diagram $K: I \to \mathcal{C}$ which has both a limit and a colimit. Recall

$$\hom(\operatorname{colim}_I K, \lim_I K) \cong \lim_{i \in I^{\mathsf{op}}} \lim_{j \in I} \hom(Ki, Kj) \cong \lim_{(i,j) \in I^{\mathsf{op}} \times I} \hom(Ki, Kj)$$

Now we know that for a general functor $F: J \to \mathsf{Set}$, we have

$$\lim F \cong \{(a_j \in Fj)_j \mid \forall \phi: i \to j.\, a_j = F(\phi) a_i\}$$

In our case we have that for $(i, i'), (j, j') \in I^{\mathsf{op}} \times I$, a morphism $(i, j) \to (i', j')$ is a pair of morphisms $\phi: i' \to i$ and $\psi: j \to j'$. The value of this functor on (ϕ, ψ) is the map $\hom(Ki, Kj) \to \hom(Ki', Kj')$ which maps $f: Ki \to Ki'$ to $K\psi \circ f \circ K\phi$. Thus we have

$$\hom(\operatorname{colim} K, \lim K) \cong \{(f_{ij}: Ki \to Kj)_{ij} \mid \forall \phi: i' \to i, \psi: j \to j'.\, f_{i'j'} = K\psi \circ f_{ij} \circ K\phi\}$$

If $\mathcal{C}$ has zero morphisms it makes sense to consider the collection $(\delta_{ij}: Ki \to Kj)_{ij}$ defined by $\delta_{ii} = \mathrm{id}_{Ki}$ and $\delta_{ij} = 0$ when $i \neq j$. Does this correspond to a morphism $\operatorname{colim} k \to \lim K$? Let us consider when $i = j, i' = j'$, then we must have

$$K\psi \circ K\phi = 1 \Longrightarrow K(\psi \circ \phi) = 1$$

for every $\phi: i' \to i, \phi: i \to i'$. So K must map endomorphisms to the identity (i.e. for $\phi: i \to i$, $K\phi = 1$). And when $i = j, i' \neq j'$ we must have $K(\psi \circ \phi) = 0$. So K must map non-endomorphisms to 0.

Such a K can be viewed as a functor from the discrete category of I (one with the same objects, but no non-identities), and as such δ is really on a morphism when the diagram is from a discrete category, i.e. the limit and colimit are products and coproducts respectively. But in such a case clearly δ is a morphism from $\coprod Ki$ to $\prod Ki$. As we saw, morphisms $\coprod Ki \to \prod Ki$ can be viewed as (potentially infinite) matrices $(f_{ij}: Ki \to Kj)$, and δ can be viewed as the identity morphism.

5.1.5 Definition

The **biproduct** of a collection of objects $\{X_i\}_{i \in I}$ is the product $\prod_i X_i$ such that $\coprod_i X_i$ also exists and the map $\delta: \coprod_i X_i \to \prod_i X_i$ is an isomorphism. A biproduct is denoted $\bigoplus_i X_i$.

The following is clear:

5.1.6 Proposition

A collection $\{X_i\}_{i \in I}$ has a biproducts iff there is an object $\bigoplus_i X_i$ along with maps $p_j: \bigoplus_i X_i \to X_j$ and $i_j: X_j \to \bigoplus_i X_i$ such that

(1) $p_k \circ i_k = 1_{X_k}$,

(2) $p_\ell \circ i_k = 0$ for $\ell \neq k$,

(3) $\bigoplus_i X_i$ equipped with the p_ks is a product of $\{X_i\}_i$,

(4) $\bigoplus_i X_i$ equipped with the i_ks is a coproduct of $\{X_i\}_i$.

5.1.7 Definition

A category is **semiadditive** if it has zero morphisms and all binary biproducts.

Suppose $\mathcal{C}$ is semiadditive, then for every $X \in \mathcal{C}$ we have canonical morphisms $\Delta\colon X \to X \oplus X$ and $\nabla\colon X \oplus X \to X$ given by the universal properties of products and coproducts respectively. That is, $p_k \circ \Delta = 1_X$ for $k = 1, 2$ and $\nabla \circ i_k = 1_X$ for $k = 1, 2$.

5.1.8 Proposition

Let $\mathcal{C}$ be semiadditive, then every hom-set forms a commutative monoid such that composition is biadditive.

Proof

Consider the operation defined by

$$\hom(X, Y) \times \hom(X, Y) \cong \hom(X, Y \oplus Y) \xrightarrow{\nabla_Y} \hom(X, Y)$$

that is, we map (f, g) to $\nabla_Y \circ (f, g)$, i.e. $f + g = \nabla(f, g)$. We claim that this gives hom-sets the structure of a commutative monoid. Firstly, this is commutative since the isomorphism is symmetric. Now consider

$$\hom(X, Y)^3 \cong \hom(X, Y \oplus Y \oplus Y) \xrightarrow{\nabla} \hom(X, Y)$$

Since $Y \oplus (Y \oplus Y) \cong (Y \oplus Y) \oplus Y$, this is just the same as both $\bullet + (\bullet + \bullet)$ and $(\bullet + \bullet) + \bullet$, i.e. addition is associative. And we have $\nabla \circ (f, 0) = \nabla \circ i_1 \circ f = f$, so 0 is the additive identity; i.e. the hom-sets indeed form commutative monoids.

Now we need to show that $(f + g) \circ h = f \circ h + g \circ h$. This is because $h \circ (f + g) = h \circ \nabla \circ (f, g)$, and $h \circ \nabla \circ i_k = h$ meaning $h \circ \nabla = \nabla(h, h)$. Thus $h \circ (f + g) = \nabla(h, h)(f, g) = \nabla(hf, hg) = hf + hg$ as desired. ■

5.2 Preadditive categories

5.2.1 Definition

A category $\mathcal{C}$ is **preadditive** if every hom-set is equipped with an operation $+$ turning it into an Abelian group, such that composition is bilinear.

A preadditive category necessarily has zero morphisms: let 0_{XY} be the additive identity in $\hom(X, Y)$. This is a zero morphism since $f \circ 0 = f \circ (0 + 0) = f \circ 0 + f \circ 0$ so $f \circ 0 = 0$, and similar for $0 \circ f$.

5.2.2 Proposition

All finite products and coproducts in a preadditive category are necessarily biproducts.

Proof

It is sufficient to show this for binary products. Suppose $X \times Y$ exists, then define $i_X \colon X \to X \times Y$ by $i_X = (1_X, 0)$ and similarly $i_Y = (0, 1_Y)$. These form a coproduct: suppose $f \colon X \to Z$ and $g \colon Y \to Z$ are morphisms then the unique morphism $\phi \colon X \times Y \to Z$ making the diagram commute is $\phi = f \circ p_X + g \circ p_Y$. Indeed, notice

$$\phi \circ (1_X, 0) = f \circ p_X(1_X, 0) + g \circ p_Y(1_X, 0) = f + g \circ 0 = f$$

And if ϕ makes this diagram commute (i.e. $\phi \circ (1, 0) = f$ and $\phi \circ (0, 1) = g$) then

$$fp_X + gp_Y = \phi((1, 0)p_X + (0, 1)p_Y)$$

and $(1, 0)p_X + (0, 1)p_Y$ is the identity (the identity of $X \times Y$ is the unique morphism such that $p_X 1 = p_X$ and $p_Y 1 = p_Y$, since morphisms are determined by their projections), so we see that $\phi = fp_X + gp_Y$ is the unique morphism satisfying this.

So we have that $X \times Y$ is a coproduct as well, and its inclusion and projection maps are related by $p_k \circ i_\ell = \delta_{k\ell}$, so it is a biproduct. ■

5.2.3 Definition

An **additive category** is a preadditive category which has all finite biproducts.

Equivalently, an additive category is a preadditive category with all finite products, or all finite coproducts, or binary products and a terminal object, etc. But importantly an additive category has binary biproducts and a zero object.

Notice that an additive category is a semiadditive category, and as such each hom-set is a commutative monoid as discussed above. The commutative monoid structure coincides with the preadditive one.

5.3 (Co)kernels, monomorphisms, epimorphisms

5.3.1 Definition

Let $\mathcal{C}$ be a category with zero morphisms. The **kernel** of a morphism $f \colon X \to Y$ is the equalizer $\ker f = \mathrm{eq}(f, 0_{XY})$, and its **cokernel** is the coequalizer $\mathrm{coker} f = \mathrm{coeq}(f, 0_{XY})$.

Kernels and cokernels need not exist. Explicitly the kernel of $f \colon X \to Y$ is a pair $(K \in \mathcal{C}, k \colon K \to X)$ such that

(1) $f \circ k = 0$

(2) For any other $h \colon Z \to X$ such that $f \circ h = 0$, there is a unique $\phi \colon Z \to K$ such that $h = k \circ \phi$.

And the cokernel of $f \colon X \to Y$ is a pair $(Q \in \mathcal{C}, q \colon Y \to Q)$ such that

(1) $q \circ f = 0$

(2) For any other $h \colon Y \to Z$ such that $h \circ f = 0$, there is a unique $\psi \colon Q \to Z$ such that $h = \psi \circ q$.

5.3.2 Example

Consider Grp, which has a zero object and is complete and cocomplete. Thus Grp has kernels and cokernels. The kernel of $f \colon G \to H$ is the usual group-theoretic kernel equipped with inclusion. The cokernel of f is the quotient of H by the normal closure of $\mathrm{im} f$, equipped with projection.

5.3.3 Proposition

Let $\mathcal{C}$ be preadditive with a zero object, then for every $f, g\colon X \to Y$, $\mathrm{eq}(f,g) = \ker(f-g)$ and $\mathrm{coeq}(f,g) = \mathrm{coker}(f-g)$.

Proof

Notice that $h \circ (f-g) = 0$ iff $h \circ f = h \circ g$. So cones over $f-g, 0$ are precisely cones over f, g. ■

5.3.4 Definition

A **monomorphism** is a morphism $f\colon X \to Y$ which is left-cancellative: $f \circ g = f \circ h$ implies $g = h$. An **epimorphism** is a morphism $f\colon X \to Y$ which is right-cancellative $g \circ f = h \circ f$ implies $g = h$.

5.3.5 Example

In Set, monomorphisms are precisely injections, and epimorphisms are precisely surjections.

5.3.6 Proposition

Let $\mathcal{C}$ be a preadditive category with a zero object, and consider a morphism $f\colon X \to Y$. Let $(K, k\colon K \to X)$ be the kernel of f and $(Q, q\colon Y \to Q)$ its cokernel.

(1) k is a monomorphism and q is an epimorphism.

(2) If $\phi\colon Y \to Z$ is a monomorphism then (K, k) is the kernel of $\phi \circ f$. Similarly if $\psi\colon Z \to X$ is an epimorphism then (Q, q) is the cokernel of $f \circ \psi$.

(3) f is a monomorphism iff K is the zero object. And f is an epimorphism iff Q is the zero object.

Proof

(1) First suppose $k \circ h = 0$. Clearly $f \circ k \circ h = 0$, and so by the universal property there is a unique morphism such that $k \circ h = k \circ g$. But clearly $g = 0$ is such a morphism, and since it is unique this means that $h = 0$. In general if $k \circ h = k \circ g$ then $k \circ (h-g) = 0$ and so $h - g = 0$ meaning $h = g$ as desired.

(2) Clearly $\phi \circ f \circ k = 0$, so now we must just show it is universal. Suppose $\phi \circ f \circ h = 0$, then $f \circ h = 0$ by monomorphism, and so by the universal property of $\ker f$, h factors through k.

(3) If f is a monomorphism then $f \circ k = 0$ implies $k = 0$. This means that K must be zero. Conversely, if $K = 0$ (and so $k = 0$), if $f \circ g = f \circ h$ then $f \circ (g-h) = 0$ and so $g - h$ must factor through 0 and as such $g - h = 0$. So $g = h$ and f is a monomorphism. ■

5.3.7 Definition

A monomorphism is **normal** if it is the kernel of some morphism. Dually an epimorphism is **conormal** if it is the cokernel of some morphism.

5.3.8 Definition

A category $\mathcal{C}$ is **Abelian** if

(1) it is preadditive with a zero object,

(2) has all binary biproducts (equivalently all finite biproducts),

(3) all kernels and cokernels exist,

(4) all monomorphisms are normal and all epimorphisms are conormal.

An Abelian category has finite biproducts, equalizers, and coequalizers. As such Abelian categories are necessarily finitely complete and finitely cocomplete.

5.3.9 Example

The category Ab is Abelian. Clearly it is preadditive, has a zero object, has binary biproducts, and kernels and cokernels exist ($\operatorname{coker} f = H/\operatorname{im}(f)$). Now we must show that monomorphisms are normal and epimorphisms are conormal.

Let $f\colon A \to B$ be a monomorphism, then consider the projection $p\colon B \to B/\operatorname{im}(f)$ (the cokernel of f). Then f is the kernel of p: clearly $p \circ f = 0$ and if $p \circ h = 0$ then we have $\operatorname{im}(h) \subseteq \ker p = \operatorname{im}(f)$. f is injective and so it has an inverse from its image $f^{-1}\colon \operatorname{im}(f) \to B$, and we have $f \circ f^{-1} \circ h = h$.

In general one does have that monomorphisms satisfy $f = \ker \operatorname{coker} f$, and epimorphisms $f = \operatorname{coker} \ker f$.

Abelian categories are important as they allow one to do homology on them. One can also discuss exact sequences in Abelian categories (to do so we must define what the image of a morphism is, which can be done in a general category, but in an Abelian category one has that it is equal to $\ker \operatorname{coker} f$).

Another interesting result (but far too complicated to prove here) is the following:

5.3.10 Theorem (Freyd-Mitchell)

All Abelian categories are isomorphic to a full subcategory of the category of modules of some ring.

That is to say, for any Abelian category $\mathcal{C}$, there is a ring R and a fully faithful embedding $\mathcal{C} \to \operatorname{Mod}_R$ (this embedding also preserves left and right exact sequences).

6 Monads

6.1 Monads from adjunctions

6.1.1 Definition

A **monad** on a category $\mathcal{C}$ is a triple (T, η, μ) where

(1) T is an endofunctor $\mathcal{C} \to \mathcal{C}$,

(2) η is the **unit**: a natural transformation $1_{\mathcal{C}} \Rightarrow T$,

(3) μ is the **multiplication**: a natural transformation $T^2 \Rightarrow T$

such that the following diagrams of functors commute

$$\begin{tikzcd} T^3 \arrow[r, "T\mu"'] \arrow[d, "\mu T"'] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"] & T \end{tikzcd} \qquad \begin{tikzcd} T \arrow[r, "\eta T"'] \arrow[dr, "1_{\mathcal{C}}"'] & T^2 \arrow[d, "\mu"] & T \arrow[l, "T\eta"] \arrow[dl, "1_{\mathcal{C}}"] \\ & T & \end{tikzcd}$$

These diagrams are reminiscient of associative and unital identities. We can make this more precise: a **monoidal category** is a category $\mathcal{C}$ equipped with a bifunctor $\otimes\colon \mathcal{C}\times\mathcal{C} \to \mathcal{C}$, a **unit** $I \in \mathcal{C}$, and various natural isomorphisms satisfying various coherence conditions (e.g. an associator $\alpha\colon A \otimes (B \otimes C) \cong (A \otimes B) \otimes C$). A **monoid object** in a monoidal category is a triple (M, μ, η) with $M \in \mathcal{C}$, $\mu\colon M \otimes M \to M$, and $\eta\colon I \to M$, satisfying certain commutativity conditions much like the above ones.

For example, if $\mathcal{C}$ has finite products then it forms a monoidal category with multiplication given by the usual category-theoretic binary product, and the unit being the terminal object. In particular a monoid in Set gives precisely the usual definition of a monoid: a set equipped with associative multiplication and an identity.

In the case of the category of endofunctors $[\mathcal{C}, \mathcal{C}]$, it forms a **strict monoidal category** in the sense that all the coherence conditions are *equalities*. Multiplication is given by composition, and the unit is $1_{\mathcal{C}}$. Indeed, associativity of functors is an equality and not just an isomorphism. A monad is precisely a monoid object in the category of endofunctors is precisely a monad over $\mathcal{C}$.

6.1.2 Lemma

Suppose we have an adjunction $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon G$ with unit $\eta\colon 1_{\mathcal{C}} \Rightarrow GF$ and counit $\epsilon\colon FG \Rightarrow 1_{\mathcal{D}}$. These form into a monad over $\mathcal{C}$: $T = GF$, the unit is η, and multiplication is given by whiskering: $\mu = G\epsilon F$.

Proof

The triangle identities give e.g. $G\epsilon F \circ \eta UF = 1_{UF}$, and so the unit diagram for the monad commutes. The multiplication diagram requires

$$(G\epsilon F)(G\epsilon FGF) = (G\epsilon F)(GFG\epsilon F)$$

both sides are just the horizontal composition $U\epsilon * U\epsilon$, and are thus equal. ■

6.1.3 Example

Consider various monads induced by free-forgetful adjunctions.

(1) The free functor $F\colon \mathsf{Set} \to \mathsf{Set}_*$ from sets to pointed sets, which adds a new base point to a set gives rise to the **maybe monad**. This monad $(\bullet)_+\colon \mathsf{Set} \to \mathsf{Set}$ adds a disjoint point to the set. The identity $\eta_A\colon A \to A_+$ is given by the natural inclusion, and multiplication $\mu_A\colon (A_+)_+ \to A_+$ restricts to the identity on A and maps the two new points in A_{++} to the unique new point in A_+. The reason for the naming will be explained later.

(2) The free functor $F\colon \mathsf{Set} \to \mathsf{Mon}$ from sets to monoids, which maps a set X to the monoid of all finite words on X gives rise to the **free monoid monad**, or the **list monad**:

$$TX = X^* := \coprod_{n \geq 0} X^n$$

The unit $\eta_X\colon X \to X^*$ is the natural inclusion, mapping $x \in X$ to the singleton list. Multiplication $\mu_X\colon X^{**} \to X^*$ is concatenation, which takes a list of lists and concatenates them to a single list.

6.1.4 Example

Other monads arise naturally:

(1) The covariant powerset functor $\mathcal{P}\colon \mathsf{Set} \to \mathsf{Set}$ (which maps a set to its powerset, and $f\colon X \to Y$ to its direct image) is a monad. The unit $\eta_A\colon A \to \mathcal{P}(A)$ maps $a \mapsto \{a\}$. Multiplication $\mu_A\colon \mathcal{P}^2(A) \to \mathcal{P}(A)$ takes a union of a set of subsets.

(2) The **Giry monad** is a monad on Meas, the category of measurable spaces and measurable functions. A measurable space is a set equipped with a σ-algebra, and a measurable function $f\colon (X, \mathcal{F}) \to (Y, \mathcal{F}')$ is a function $f\colon X \to Y$ such that the inverse image f^{-1} maps $\mathcal{F}'$ into $\mathcal{F}$: $f^{-1}(Y') \in \mathcal{F}$ for $Y' \in \mathcal{F}'$.

The Giry monad maps X to $\mathrm{Prob}(X)$: the space of probability measures on X, equipped with the smallest σ-algebra making every $\mathrm{ev}_A\colon \mathrm{Prob}(X) \to [0,1]$ (mapping μ to $\mu(A)$) measurable for $A \subseteq X$ measurable. The unit $\eta_X\colon X \to \mathrm{Prob}(X)$ maps $x \in X$ to the Dirac measure δ_x, which maps $A \subseteq X$ to 1 iff $x \in A$ and 0 otherwise. Multiplication $\mu_X\colon \mathrm{Prob}^2(X) \to \mathrm{Prob}(X)$ is defined via integration.

We can dualize the notion of a monad:

6.1.5 Definition

A **comonad** on $\mathcal{C}$ is a monad on $\mathcal{C}^{\mathsf{op}}$. Explicitly, it is an endofunctor $K\colon \mathcal{C} \to \mathcal{C}$ with counit $\epsilon\colon K \Rightarrow 1_{\mathcal{C}}$, and comultiplication $\delta\colon K \Rightarrow K^2$ such that the dual diagrams for monads commute.

A comonad is a comonoid in the category of endofunctors of $\mathcal{C}$. The dual of the above lemma (recall $F \dashv G$ iff $G^{\mathsf{op}} \dashv F^{\mathsf{op}}$) tells us that comonads arise from the right adjoint of an adjunction.

6.1.6 Example

A monad on a preorder $(P, \leq)$ is an order-preserving function $T\colon P \to P$ such that $p \leq Tp$ and $T^2p \leq Tp$. If P is a poset (i.e. antisymmetric), then these imply $T^2p = Tp$. An order-preserving function such that $p \leq Tp$ and $T^2 = T$ is called a **closure operator**.

A comonad is an order preserving function $K\colon P \to P$ such that $Kp \leq p$ and $Kp \leq K^2p$. When P is a poset, this again implies $K^2 = K$ and $Kp \leq p$, which makes K into a **kernel operator**.

For example, if X is a topological space, then the poset $(\mathcal{P}X, \subseteq)$ admits both a closure and kernel operator. Closure maps a subset A to its closure (the smallest closed set containing A), and the kernel operator maps a subset A to its interior (the largest open set contained in A).

6.2 Adjunctions from monads

In the previous section we showed that adjunctions give rise to monads. In this section we will show the converse is true: monads give rise to adjunctions.

6.2.1 Definition

Let $\mathcal{C}$ be a category with a monad (T, η, μ). The **Eilenberg-Moore category** for T, or the **category of T-algebras** is the category $\mathcal{C}^T$ defined by:

(1) Objects are pairs $(A \in \mathcal{C}, a\colon TA \to A)$ so that $a\eta_A = 1_A$ and $a \circ \mu_A = a \circ Ta$, i.e. the following diagrams commute:

$$\begin{tikzcd} A \arrow[r, "\eta_A"'] \arrow[dr, "1_A"'] & TA \arrow[d, "a"] \\ & A \end{tikzcd} \qquad \begin{tikzcd} T^2A \arrow[r, "\mu_A"'] \arrow[d, "Ta"'] & TA \arrow[d, "a"] \\ TA \arrow[r, "a"] & A \end{tikzcd}$$

(2) Morphisms $f\colon (A,a) \to (B,b)$ are T-algebra **homomorphisms**: maps $f\colon A \to B$ in $\mathcal{C}$ so that $b \circ Tf = f \circ a$:

$$\begin{tikzcd} TA \arrow[r, "Tf"'] \arrow[d, "a"'] & TB \arrow[d, "b"] \\ A \arrow[r, "f"] & B \end{tikzcd}$$

Composition in $\mathcal{C}^T$ is given via composition in $\mathcal{C}$, and identities are the identities in $\mathcal{C}$.

6.2.2 Example

(1) Consider the monad $\bullet_+\colon \mathsf{Set} \to \mathsf{Set}$. An algebra here is a set A with a function $a\colon A_+ \to A$ such that $a\eta_A = 1_A$, i.e. a restricts to the identity on A and $a\mu_A = aa_+$ which adds no more information (since a_+ maps the new point in A_{++} to the point in A_+, as does μ_A). Thus the data of an algebra is just the choice of some point $a \in A$.

Then a homomorphism $f\colon (A,a) \to (B,b)$ is just a function $f\colon A \to B$ such that $f(a) = b$. Thus the Eilenberg-Moore category is isomorphic to Set_*, the category of pointed sets.

(2) Consider the free monoid monad (list monad) $\bullet^*\colon \mathsf{Set} \to \mathsf{Set}$. An algebra is a set A with a function $a\colon A^* \to A$, so a collection of functions $(a_n\colon A^n \to A)_{n\geq 0}$ such that

(i) $a\eta_A = 1_A$, meaning $a_1 = 1_A$;

(ii) $a\mu_A = aa^*$, which says that given a list of lists $((x_{11},\dots,x_{1m_1}),\dots,(x_{n1},\dots,x_{nm_n}))$, the following are equal:

$$a(x_{11},\dots,x_{1m_1},\dots,x_{n1},\dots,x_{nm_n}) = (a(x_{11},\dots,x_{1m_1}),\dots,a(x_{n1},\dots,x_{nm_n}))$$

Now recall that a monoid is a set A with an associative operation $A^2 \to A$ and a unit $e \in A$. A monoid defines an algebra: the 0-ary operation (a_0) picks out e, a_1 is the identity, a_2 is the binary operation. Then the n-ary operation is given by associativity: map $(x_1,\dots,x_n)$ to the image of $(x_1,(x_2,(\dots)))$. Conversely an algebra defines a monoid via a_0, a_2.

Algebra morphisms correspond to monoid morphisms, so there is an equivalence of algebras and monoids.

For a monad T, there is an evident forgetful functor $U^T\colon \mathcal{C}^T \to \mathcal{C}$.

6.2.3 Lemma

Let (T,η,μ) be a monad on $\mathcal{C}$. Then U^T is right-adjoint to a free functor $F^T\colon \mathcal{C} \to \mathcal{C}^T$ such that the monad induced by the adjunction is (T,η,μ).

Proof

The free functor $F^T\colon \mathcal{C} \to \mathcal{C}^T$ maps $A \in \mathcal{C}$ to the **free algebra**: $F^T A = (TA, \mu_A\colon T^2A \to TA)$, and maps $f\colon A \to B$ to the morphism $F^T f\colon (TA, \mu_A) \xrightarrow{Tf} (TB, \mu_B)$. This is indeed a well-defined morphism: we require $F^T f \circ \mu_A = \mu_B \circ TF^T f$, which is just $TF \circ \mu_A = \mu_B \circ T^2 f$; precisely the naturality condition of multiplication.

Note that $U^T F^T = T$. So let us take the unit of the adjunction to be η, and define the counit $\epsilon\colon F^T U^T \Rightarrow 1_{\mathcal{C}^T}$ whose component at (A, a) is

$$\epsilon_{(A,a)}\colon (TA, \mu_A) \xrightarrow{a} (A, a)$$

That is, the counit at (A, a) is the algebra homomorphism a. Note that $U^T \epsilon F^T = \mu$, since $U^T \epsilon_{F^T(A)} = U^T \epsilon_{(TA,\mu_A)} = \mu_A$. Thus assuming η, ϵ do define a unit-counit pair, we have finished.

All that remains is to show the triangle identities: $(\epsilon F^T)(F^T \eta)$ and $(U^T \epsilon)(\eta U^T)$ are the identities. Indeed:

$$(\epsilon F^T)(F^T \eta)_A = \epsilon_{(TA,\mu_A)} \circ T\eta_A = \mu_A \circ T\eta_A = 1_A$$

by the definition of a monad, and

$$(U^T \epsilon)(\eta U^T)_{(A,a)} = a \circ \eta_a = 1_A$$

by the definition of a homomorphism. ■

6.2.4 Definition

Let (T, η, μ) be a monad on $\mathcal{C}$. The **Kleisli category** $\mathcal{C}_T$ is defined so that

(1) its objects are the objects of $\mathcal{C}$,

(2) a morphism $A \to B$ in $\mathcal{C}^T$ is a morphism $A \to TB$ in $\mathcal{C}$: $\hom_{\mathcal{C}^T}(A, B) = \hom_{\mathcal{C}}(A, TB)$.

Identities and composition are defined as follows:

(1) The unit $\eta_A\colon A \to TA$ is the identity of A in $\mathcal{C}_T$,

(2) The composition of $f\colon A \to TB$ with $g\colon B \to TC$ is the composite

$$A \xrightarrow{f} TB \xrightarrow{Tg} T^2C \xrightarrow{\mu_C} TC$$

A morphism from A to B in $\mathcal{C}_T$ is depicted $A \rightsquigarrow B$.

Composition is indeed associative: given $f\colon A \to TB, g\colon B \to TC, h\colon C \to TD$, the two composites are:

$$h \circ (g \circ f)\colon A \xrightarrow{f} TB \xrightarrow{Tg} T^2C \xrightarrow{\mu_C} TC \xrightarrow{Th} T^2D \xrightarrow{\mu_D} TD$$

$$(h \circ g) \circ f\colon A \xrightarrow{f} TB \xrightarrow{Tg} T^2C \xrightarrow{T^2h} T^3D \xrightarrow{T\mu_D} T^2D \xrightarrow{\mu_D} TD$$

So we need to show the equality $\mu_D \circ T\mu_D \circ T^2h = \mu_D \circ Th \circ \mu_C$. By naturality of μ we have $\mu_{TD} \circ T^2h = Th \circ \mu_C$, so it is sufficient to show $\mu_D \circ \mu_{TD} \circ T^2h = \mu_D \circ T\mu_D \circ T^2h$. This is precisely due to μ being multiplicative: $\mu \circ T\mu = \mu \circ \mu T$.

And η does indeed form an identity: given $f\colon A \to TB$, $f \circ \eta_A$ is

$$A \xrightarrow{\eta_A} TA \xrightarrow{Tf} T^2A \xrightarrow{\mu_A} TA$$

by naturality of η, $Tf \circ \eta_A = \eta_{TA} \circ f$ and $\mu_A \circ \eta_{TA} = 1_A$ as desired. And $\eta_B \circ f$ is

$$A \xrightarrow{f} TB \xrightarrow{T\eta_B} T^2B \xrightarrow{\mu_B} TB$$

which is again just f. So $\mathcal{C}_T$ is indeed a category.

6.2.5 Example

(1) Consider the maybe monad $T = \bullet_+$. A map $A \rightsquigarrow B$ is $A \to B_+$ which can be considered partial maps from A to B. The composition of $A \rightsquigarrow B \rightsquigarrow C$ is simply the composition of partial maps. Thus $\mathcal{C}_T$ is equivalent to Set^∂, the category of sets with partially-defined functions.

(2) Consider the list monad $T = \bullet^*$. A map $A \rightsquigarrow B$ is $A \to B^*$, and a composition $g \circ f$ maps $a \in A$ to a list $f(a) \in B^*$ then applys g to each element of the list and concatenates them.

(3) Let S be a fixed set of "states", and consider the adjunction $S \times \bullet \dashv \bullet^S$. This induces a monad $(S \times \bullet)^S$ on Set. A $A \rightsquigarrow B$ in the Kleisli category is, by adjunction, a map $f\colon A \times S \to B \times S$. That is, it is a map which maps an input $a \in A$ and a state $s \in S$, and returns a new value $f_s(a) \in B$ along with a new state $s' \in S$. This monad is called the **side-effect monad**.

(4) The Kleisli category of the Giry monad is the category of measurable spaces and **Markov kernels**.

6.2.6 Lemma

For any monad (T, η, μ) on $\mathcal{C}$, there is an adjunction $F_T\colon \mathcal{C} \rightleftarrows \mathcal{C}_T\colon U_T$ whose induced monad is (T, η, μ).

Proof

The functor F_T is the identity on objects and maps $f\colon A \to B$ to the morphism $A \rightsquigarrow B$ defined by

$$F_T f\colon A \xrightarrow{f} B \xrightarrow{\eta_B} TB$$

And the functor U_T maps $A \in \mathcal{C}_T$ to $TA \in \mathcal{C}$ and a morphism $g\colon A \rightsquigarrow B$ to the morphism

$$U_T g\colon TA \xrightarrow{Tg} T^2B \xrightarrow{\mu_B} TB$$

The functorality of these definitions is readily provable. Their adjunction is simple, as

$$\mathcal{C}_T(F_T A, B) = \mathcal{C}_T(A, B) = \mathcal{C}(A, TB) = \mathcal{C}(A, U_T B)$$

and naturality is again readily provable. The unit of this adjunction is the image of the identity in $\mathcal{C}_T$, which is just η. The counit is the image of the identity of $TA \to TA$, i.e. $TA \rightsquigarrow A$. ■

6.2.7 Definition

Let (T, η, μ) be a monad on $\mathcal{C}$. Define the category Adj_T whose objects are fully-specified adjunctions $(F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon U, \eta, \epsilon)$ inducing the monad T on $\mathcal{C}$. A morphism from $F \dashv U$ to $F' \dashv U'$ is a functor $K\colon \mathcal{D} \to \mathcal{D}'$ such that $KF = F'$ and $U'K = U$ (i.e. the obvious diagram commutes).

Recall that a morphism of adjunctions $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon G$ to $F'\colon \mathcal{C}' \rightleftarrows \mathcal{D}'\colon G'$ is a pair of functors $H\colon \mathcal{C} \to \mathcal{C}'$ and $K\colon \mathcal{D} \to \mathcal{D}'$ such that $KF = F'H$ and $HG = G'K$ and one of the following equivalent conditions holds:

(1) $H\eta = \eta' H$ (η, η' are the respective units of the adjunction), or

(2) $K\epsilon = \epsilon' K$, or

(3) Transposition commutes with H, K, i.e. the following diagram commutes:

$$\begin{tikzcd}
\mathcal{D}(Fc,d) \arrow[r, "\cong"'] \arrow[d, "K"'] & \mathcal{C}(c,Gd) \arrow[d, "H"] \\
\mathcal{D}'(KFc,Kd) \arrow[d, "="'] & \mathcal{C}'(Hc,HGd) \arrow[d, "="] \\
\mathcal{D}'(F'Hc,Kd) \arrow[r, "\cong"] & \mathcal{C}'(Hc,G'Kd)
\end{tikzcd}$$

Note that since adjunctions in Adj_T have the same unit (η), the first of the above conditions hold ($H = 1_{\mathcal{C}}$). Thus Adj_T is a subcategory of the category of adjunctions. In particular this means that K satisfies the other two conditions: $K\epsilon = \epsilon' K$ and given $f\colon c \to Ud = U'Kd$, transposing Kf is the same as the tranpose of f.

6.2.8 Proposition

The Kleisli category $\mathcal{C}_T$ is initial in Adj_T, and the Eilenberg-Moore category $\mathcal{C}^T$ is terminal.

Proof

Suppose we have an adjunction $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon U$ with induced monad T. The requirement of a monadic morphism J from $\mathcal{C}_T$ to $\mathcal{D}$ requires that $JF_T = F$ and $UJ = U_T$. Since F_T is the identity on objects, this requires us to define $Jc = Fc$ on objects $c \in \mathcal{C}_T$.

Furthermore, let $\epsilon\colon FU \Rightarrow 1_{\mathcal{D}}$ be the counit of $F \dashv U$. For J to be a monadic morphism, we must have that J commutes with transposition: if we take a morphism $f\colon c \rightsquigarrow c'$ in $\mathcal{C}_T$ and then transpose it via $F_T \dashv U_T$ to get $f\colon c \to Tc' = U_T c'$, and then transpose it via $F \dashv U$, it is the same as Jf. The transposition of $f\colon c \to U_T c'$ is (recall $T = UF$),

$$Jf\colon Fc \xrightarrow{Ff} FTc' = FUFc' \xrightarrow{\epsilon F_{c'}} Fc'$$

This is readily shown to be functorial (remember $U\epsilon F = \mu$).

Similarly to define $K\colon \mathcal{D} \to \mathcal{C}^T$, we must have $U^T K = U$ so that Kd is a suitable T-algebra on Ud. For an algebra $(c, \gamma\colon Tc \to c) \in \mathcal{C}^T$, we have that $\epsilon^T_{(c,\gamma)} = \gamma$ for the counit of $F^T \dashv U^T$. So γ can be viewed as a morphism $\gamma\colon (Tc, \mu_c) \to (c, \gamma)$ from the free T-algebra on c to the given algebra, and it is the transpose of the identity on $c = U^T(c, \gamma)$. K needs to preserve transposes, and thus we need to define $Kd = (Ud, U\epsilon_d)$; $U\epsilon_d$ being a homomorphism is due to $U\epsilon F = \mu$. Since $U^T K = U$, we have that $Kf = Uf$. Functorality is then obvious. ■

Recall that a **free T-algebra** is one of the form

$$F^T c = (Tc, \mu_c) \in \mathcal{C}^T$$

6.2.9 Lemma

The unique functor $K\colon \mathcal{C}_T \to \mathcal{C}^T$ is fully faithful, and its image are all the free T-algebras.

Proof

Our definition above tells us $Kc = (U_T c, U_T \epsilon_c)$; $U_T c = Tc$ and recall that ϵ_c is the identity of $\mathcal{C}$ on Tc and thus $U_T \epsilon_c = \mu_c$. So $Kc = (Tc, \mu_c)$, i.e. its image are the free T-algebras. The action of K,

$$\mathcal{C}_T(c, c') \xrightarrow{K} \mathcal{C}^T((Tc, \mu_c), (Tc', \mu_{c'}))$$

commutes with the transposition isomorphisms:

$$\begin{tikzcd} \mathcal{C}_T(c,c') \arrow[r, "\cong", "\cong"'] \arrow[d, "K"] & \mathcal{C}(c,Tc') \arrow[d, "="] \\ \mathcal{C}^T((Tc,\mu_c),(Tc',\mu_{c'})) & \mathcal{C}(c,Tc') \end{tikzcd}$$

So K is a composition of isomorphisms and thus an isomorphism, as desired. ■

6.3 Monadic functors

Consider the free-forgetful adjunction $\mathbb{Z}[\bullet]\colon \mathsf{Set} \rightleftarrows \mathsf{Ab}\colon U$; $\mathbb{Z}[S]$ is the set of all formal $\mathbb{Z}$-linear combinations of elements of S. For $f\colon S \to T$, $\mathbb{Z}[f]\colon \mathbb{Z}[S] \to \mathbb{Z}[T]$ maps $\sum_i n_i s_i$ to $\sum_i n_i f(s_i)$. The unit of this adjunction is $\eta_S\colon S \to U\mathbb{Z}[S]$, which maps $s \in S$ to the formal sum consisting of only s, i.e. the natural inclusion $S \subseteq \mathbb{Z}[S]$. The counit is $\epsilon_A\colon \mathbb{Z}[UA] \to A$, which "evaluates" a formal sum $\sum_i n_i a_i \in \mathbb{Z}[UA]$. Thus multiplication is $\mu = U\epsilon\mathbb{Z}[\bullet]$, which is the map $\mu_S\colon \mathbb{Z}[\mathbb{Z}[S]] \to \mathbb{Z}[S]$ evaluating a sum of sums.

By our above consideration, there is a unique functor $K\colon \mathsf{Ab} \to \mathsf{Set}^{\mathbb{Z}[\bullet]}$ commuting with the aforementioned adjunction and $F^{\mathbb{Z}[\bullet]} \dashv U^{\mathbb{Z}[\bullet]}$. Our definition gives us

$$K(A) = (UA, U\epsilon_A)$$

That is, $K(A)$ is the algebra over A whose map is the evaluation map, $\mathrm{ev}_A\colon \mathbb{Z}[A] \to A$. Now, K is actually an isomorphism of categories.

Indeed, an object $(A, a\colon \mathbb{Z}[A] \to A)$ in the Eilenberg-Moore category is a set A and a morphism a such that $1_A = a\eta_A$ so that a is the identity on $A \subseteq \mathbb{Z}[A]$, and $a\mu_A = a\mathbb{Z}[a]$. This a gives A the structure of an Abelian group: where we define $x + y$ to be $a(x + y)$ for $x, y \in A$. And a morphism $f\colon (A, a) \to (B, b)$ is just a function $f\colon A \to B$ which preserves evaluations of formal sums, i.e. a group homomorphism.

This is to say, $\mathbb{Z}[\bullet] \dashv U$ is a *monadic adjunction*:

6.3.1 Definition

An adjunction $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon U$ is **monadic** if the unique adjunction functor $\mathcal{D} \to \mathcal{C}^T$ is an equivalence of categories. A functor $U\colon \mathcal{D} \to \mathcal{C}$ is **monadic** if it has a left adjoint making a monadic adjunction.

An adjunction or functor is **strictly monadic** if the unique adjunction functor is an isomorphism instead of just an equivalence.

Recall that a reflective subcategory $\mathcal{D} \hookrightarrow \mathcal{C}$ is one where the inclusion has a left adjoint (a **reflector**) $L\colon \mathcal{C} \to \mathcal{D}$. The induced endofunctor $L\colon \mathcal{C} \to \mathcal{C}$ defines a monad with unit $\eta_C\colon C \to LC$. Multiplication is $\mu_C = \mathrm{inc}\epsilon L$, which we note is an isomorphism (assuming $\mathcal{D}$ is a full subcategory). A monad whose multiplication is an isomorphism is called an **idempotent monad**.

6.3.2 Proposition

The inclusion of a reflective subcategory is monadic.

Proof

An L-algebra is an object $C \in \mathcal{C}$ along with $c\colon LC \to C$ such that $c\eta_C = 1_C$ and $c\mu_C = c \circ Lc$. By naturality, $\eta_C c = Lc\eta_{LC}$; and $\eta_{LC} = L\eta_C$ as both are left inverses of the isomorphism μ_C. So $\eta_C c = L(c\eta_C) = 1_{LC}$; meaning c and η_C are inverse isomorphisms. Furthermore $\mu_C = Lc$ is then satisfied since both are inverses to $\eta_{LC} = L\eta_C$. So if $C \in \mathcal{C}$ admits an L-algebra, η_C must be an isomorphism; and conversely $\eta_C^{-1}\colon LC \to C$ forms C into an L-algebra. Thus $\mathcal{C}^L$ has as objects (C, η_C^{-1}).

And it can be shown that $C \in \mathcal{C}$ is in the essential image of the inclusion of $\mathcal{D}$ iff η_C is an isomorphism.

Thus all elements of $\mathcal{C}^L$ are isomorphic to (the unique) algebras over elements of $\mathcal{D}$, meaning K is essentially surjective. Full faithfulness is due to naturality of η. ■

6.4 Canonical presentations via free algebras

6.4.1 Definition

A **split coequalizer** diagram is one of the form

$$\begin{tikzcd} x \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & y \arrow[l, bend left, "t"] \arrow[r, "h"] & z \arrow[l, bend left, "s"] \end{tikzcd}$$

So that

$$hf = hg, \qquad hs = 1_z, \qquad gt = 1_y, \qquad ft = sh$$

The condition $hf = hg$ says that this triple f, g, h defines a **fork**: h is a cone under $f, g\colon x \to y$.

6.4.2 Lemma

The underlying fork of a split coequalizer diagram is a coequalizer. Moreso, it is an **absolute colimit**: it is preserved by any functor.

Proof

Let $k\colon y \to w$ so that $kf = kg$; we must show k factors uniquely through h. That is, there is a $\ell\colon z \to w$ such that $k = \ell h$. Note that h being a split epimorphism means that ℓ is necessarily unique: $\ell hs = \ell$. The factorization is just $\ell = ks$:

$$ksh = kft = kgt = k$$

A split coequalizer diagram is clearly preserved under any functor. That is, the diagram Ff, Fg, Fh is another split coequalizer diagram and thus is a coequalizer. ■

6.4.3 Example

Let $(A, \alpha\colon TA \to A)$ be a T-algebra. Then the following is a split coequalizer diagram:

$$\begin{tikzcd} T^2A \arrow[r, shift left, "T\alpha"] \arrow[r, shift right] & TA \arrow[l, bend left, "\mu_A"] \arrow[r, "\alpha"] & A \arrow[l, bend left, "\eta_A"] \end{tikzcd}$$

$$\eta_{TA}$$

6.4.4 Definition

Given a functor $U\colon \mathcal{D} \to \mathcal{C}$,

(1) A **U-split coequalizer** is a parallel pair $f, g\colon x \to y$ with a split coequalizer diagram

$$\begin{tikzcd} Ux \arrow[r, shift left, "Uf"] \arrow[r, shift right, "Ug"'] & Uy \arrow[l, bend left, "t"] \arrow[r, "h"] & z \arrow[l, bend left, "s"] \end{tikzcd}$$

(2) U **creates coequalizer of U-split pairs** if any U-split coequalizer admits a coequalizer in $\mathcal{D}$ whose image under U is isomorphic to the underlying fork of the given U-split coequalizer diagram,

and if any such fork (whose image under U is isomorphic to the fork of the U-split diagram) is a coequalizer.

(3) U **strictly creates coeqalizers of** U**-split pairs** if any U-split coequalizer admits a unique lift to a coequalizer in $\mathcal{D}$ (for the given parallel pair).

6.4.5 Proposition

Let T be a monad on $\mathcal{C}$. Then the monadic forgetful functor $U^T\colon \mathcal{C}^T \to \mathcal{C}$ strictly creates coequalizers of U^T-split pairs.

Proof

Suppose we have a parallel pair $f, g\colon (A, \alpha) \to (B, \beta)$ in $\mathcal{C}^T$ with a U^T-splitting:

$$\begin{tikzcd} x \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & y \arrow[l, bend left, "t"] \arrow[r, "h"] & z \arrow[l, bend left, "s"] \end{tikzcd}$$

We must show that C lifts to an algebra (C, γ) and h lifts to an algebra map such that they form a coequalizer for f, g in $\mathcal{C}^T$, and that such a lift is unique.

Recall that split coequalizers are absolute colimits, so that Th is the coequalizer of Tf and Tg. As f, g are algebra morphisms, we have that $\beta Tf = f\alpha$ and $\beta Tg = g\alpha$. In particular

$$h\beta Tf = hf\alpha = hg\alpha = h\beta Tg$$

and by the universal property of the colimit, there must be a unique map $\gamma\colon TC \to C$ which makes the following diagram commute:

$$\begin{tikzcd} TA \arrow[r, shift left, "Tf"] \arrow[r, shift right, "Tg"'] \arrow[d, "\alpha"] & TB \arrow[r, "Th"] \arrow[d, "\beta"] & TC \arrow[d, "\gamma"] \\ A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \arrow[r, "h"] & C \end{tikzcd}$$

In particular $h\colon (B, \beta) \to (C, \gamma)$ is an algebra morphism, assuming that (C, γ) forms a T-algebra. Since γ is the unique morphism making the above square commute, (C, γ) is the unique lift of C such that h is an algebra map. We have that

$$\gamma\eta_C h = \gamma Th\eta_B = h\beta\eta_B = h$$

(the first equality is due to the naturality of η, the second due to the commuting square above.) Since h is epic (right-cancellative) as a coequalizer, $\gamma\eta_C = 1_C$ as desired. Similarly

$$\gamma\mu_C T^2 h = \gamma Th\mu_B = h\beta\mu_B = h\beta T\beta = \gamma ThT\beta = \gamma T\gamma T^2 h$$

(diagram chasing on a 3d diagram.) The split coequalizer is preserved by T^2, in particular $T^2 h$ is a coequalizer and thus epic, so we have $\gamma\mu_C = \gamma T\gamma$, as desired. Meaning (C, γ) is indeed a T-algebra.

Now h is a coequalizer in $\mathcal{C}^T$. Suppose $k\colon (B, \beta) \to (D, \delta)$ satisfies $kf = kg$; then we can factorize k through a unique $j\colon C \to D$. j defines a T-algebra morphism: we must show $j\gamma = \delta Tj$. Indeed

$$j\gamma Th = jh\beta = k\beta = \delta Tk = \delta TjTh$$

and by Th's epicness, $j\gamma = \delta Tj$ as desired. ■

6.4.6 Corollary

Let T be a monad on $\mathcal{C}$, and $(A, \alpha\colon TA \to A)$ a T-algebra. Then

$$\begin{tikzcd} (T^2A, \mu_{TA}) \arrow[r, shift left, "T\alpha"] & (TA, \mu_A) \arrow[l, shift left, bend left, "\mu_A"] \arrow[r, "\alpha"] & (A, \alpha) \arrow[l, bend left, "\eta_A"] \end{tikzcd}$$

$$\eta_{TA}$$

is a coequalizer diagram in $\mathcal{C}^T$.

Proof

As already noted in the above example, the image of this diagram under U^T is a split coequalizer diagram. We just showed U^T strictly creates U^T-split coequalizers; as this given diagram in $\mathcal{C}^T$ is the unique lift of the U^T-split diagram, it must be a coequalizer. ■

6.4.7 Corollary

Let $F\colon \mathcal{C} \rightleftarrows \mathcal{D}\colon U$ be a monadic adjunction, then

(1) $U\colon \mathcal{D} \to \mathcal{C}$ creates coequalizers of U-split pairs,

(2) For any $D \in \mathcal{D}$, there is a coequalizer diagram

$$\begin{tikzcd} FUFU(D) \arrow[r, shift left, "FU\epsilon_D"] \arrow[r, shift right, "\epsilon_{FUD}"'] & FUD \arrow[r, "\epsilon_D"] & D \end{tikzcd}$$

Proof

$U\colon \mathcal{D} \to \mathcal{C}$ being monadic means that there is an equivalence of categories $K\colon \mathcal{D} \to \mathcal{C}^T$, where T is the monad UF, and $U = U^T K$ and $F^T = KF$.

To show the first point, suppose $f, g\colon A \rightrightarrows B$ is U-split in $\mathcal{D}$. Since $U = U^T K$, we have that $Kf, Kg\colon KA \rightrightarrows KB$ is U^T-split. Then we can uniquely lift the fork of the U^T-split coequalizer diagram to a coequalizer of Kf, Kg, and then via an inverse equivalence $L\colon \mathcal{C}^T \to \mathcal{D}$, lift it to a coequalizer of f, g in $\mathcal{D}$. U also reflects coequalizers which are U-split, since U^T reflects U^T-split coequalizers and K reflects all coequalizers. Thus if a fork is mapped to one isomorphic to a U-split diagram, it must be a coequalizer. All in all, U creates coequalizers of U-split pairs.

The image under U of the diagram is split (take $t = \eta_{UFUD}$ and $s = \eta_{UD}$), and since U reflects U-split coequalizers this means the original diagram is a coequalizer. ■

6.5 Recognizing categories of algebras

6.5.1 Theorem

A right adjoint $U\colon \mathcal{D} \to \mathcal{C}$ is monadic iff it creates coequalizers of U-split pairs. It is strictly monadic iff it strictly creates coequalizers of U-split pairs.

Proof

We only prove the first statement. Let F be a left adjoint, and consider the unique map $K\colon \mathcal{D} \to \mathcal{C}^T$. We need to show that K is an equivalence iff U creates coequalizers of U-split pairs. Our previous proposition

already proved the first direction; monadic functors create coequalizers of their own split pairs. So now suppose U creates coequalizers of U-split pairs.

We have $U^T K = U$ and $KF = F^T$, so if $L\colon \mathcal{C}^T \to \mathcal{D}$ is an inverse equivalence then we must have $U^T \cong UL$ and $F \cong LF^T$; we will use these conditions to guide our definition of L. For free algebras, we have this isomorphism hold strictly:

$$L(TA, \mu_A) = FA$$

and for the free map $Tf\colon (TA, \mu_A) \to (TB, \mu_B)$ map it to $LTf = Ff$. We now need to expand this to non-free algebras and maps.

Equivalences preserve limits and colimits. For an algebra $(A, \alpha) \in \mathcal{C}^T$, consider the parallel morphisms $F\alpha, \epsilon_{FA}\colon FUFA \rightrightarrows FA$. Applying U gives us $T\alpha, \mu_A\colon T^2A \rightrightarrows TA$, which we showed is split; thus $F\alpha, \epsilon_{FA}$ is U-split and so has a coequalizer. Define $L(A, \alpha)$ to be this coequalizer:

$$FUFA \underset{\epsilon_{FA}}{\overset{F\alpha}{\rightrightarrows}} FA \to L(A, \alpha)$$

For a map $f\colon (A, \alpha) \to (B, \beta)$ note that by naturality of ϵ, $Ff \circ \epsilon_{FA} = \epsilon_{FB} \circ FUFf$. And so there is a unique map Lf making the following diagram commute:

$$\begin{tikzcd}
FUFA \arrow[r, shift left, "F\alpha"] \arrow[r, shift right, "\epsilon_{FA}"'] \arrow[d, "FUFf"'] & FA \arrow[r, two heads] \arrow[d, "Ff"] & L(A, \alpha) \arrow[d, "Lf"] \\
FUFB \arrow[r, shift left, "F\beta"] \arrow[r, shift right, "\epsilon_{FB}"'] & FB \arrow[r, two heads] & L(B, \beta)
\end{tikzcd}$$

In other words, we define L to be the composite of $\mathcal{C}^T \to \mathcal{D}^{\bullet\rightrightarrows\bullet} \xrightarrow{\text{colim}} \mathcal{D}$; this is functorial.

Recall that $K\colon \mathcal{D} \to \mathcal{C}^T$ is defined by $K(D) = (UD, U\epsilon_D)$ and $Kf = Uf$, thus K maps

$$FUFA \underset{\epsilon_{FA}}{\overset{F\alpha}{\rightrightarrows}} FA \to L(A, \alpha) \mapsto \ K\ (T^2A, \mu_{TA}) \underset{\mu_A}{\overset{T\alpha}{\rightrightarrows}} (TA, \mu_A) \to KL(A, \alpha)$$

U^T carries the parallel pair to a U^T-split parallel pair, and since it strictly creates such coequalizers, $KL(A, \alpha)$ must be the coequalizer of this diagram; and we already know this is (A, α). Thus $KL \cong 1_{\mathcal{C}^T}$.

We must show that $LK \cong 1_{\mathcal{D}}$ as well. Given $D \in \mathcal{D}$, LKD is the coequalizer of

$$FUFUD \underset{\epsilon_{FUD}}{\overset{FU\epsilon_D}{\rightrightarrows}} FUD$$

which we already know to be D. Thus $LKD \cong D$ uniquely so that it commutes with the coequalizer diagrams, and such an isomorphism must be natural $LD \cong 1_{\mathcal{D}}$ as desired. ■

6.5.2 Example

The free-forgeful adjunction $F\colon \mathsf{Set} \rightleftarrows \mathsf{Monoid}\colon U$ is monadic. It is sufficient to show that U strictly creates the coequalizer of any pair of monoid homomorphisms $f, g\colon M \to M'$ whose underlying functions extend to a split coequalizer diagram in Set. So we have

$$M \underset{g}{\overset{f}{\rightrightarrows}} M' \xrightarrow{h} M''$$

which split via t, s; we need to give M'' a monoid structure making $h\colon M' \to M''$ a monoid morphism in such a way to make h a coequalizer of f, g.

For h to be a homomorphism, M'''s unit must be the composite $\eta''\colon 1 \xrightarrow{\eta'} M' \xrightarrow{h} M''$. To define multiplication for M'', that is $\mu''\colon M'' \times M'' \to M''$, we use the fact that split coequalizers are absolute. In particular, we use the functor $\bullet^2\colon \mathsf{Set} \to \mathsf{Set}$, yielding parallel equalizer diagrams and a unique map μ'' making it all commute:

$$\begin{tikzcd}
M \times M \arrow[r, "f \times f, g \times g"'] \arrow[d, "\mu"] & M' \times M' \arrow[r, "h \times h"'] \arrow[d, "\mu'"] & M'' \times M'' \arrow[d, "\mu''"] \\
M \arrow[r, "f, g"] & M' \arrow[r, "h"] & M''
\end{tikzcd}$$

Assuming that (M'', η'', μ'') do indeed form a monoid, we are finished; this is the unique lift of h and h must be a coequalizer. The proof that this is indeed a monoid proceeds similar to before; we follow diagrams and quotient by an epic. ■

6.5.3 Example

The free-forgetful adjunctions from Set to $\mathsf{Grp}, \mathsf{Ab}, \mathsf{Ring}$ are all monadic (the free ring on an Abelian group A is $\bigoplus_{n \geq 0} A^n$, multiplication given by concatenation; thus the free ring on a set S is $\bigoplus_{n \geq 0} \mathbb{Z}[S]^n$).

6.5.4 Definition

A **filtered category** is a small category I such that

(1) every two objects $x, y \in I$ have a common bound: there is an object $z \in I$ and morphisms $x \to z \leftarrow y$;

(2) for every two parallel morphisms $u, v \colon x \rightrightarrows y$ there is a morphism $w \colon y \to z$ such that $wu = wv$.

A **filtered colimit** is the limit of a functor $I \to \mathcal{C}$ for I filtered.

A functor is **finitary** if it preserves filtered colimits.

If we have an adjunction $F \dashv U$ such that U is finitary, so is the induced monad $T = UF$ as F preserves colimits in general.

6.5.5 Definition

A category $\mathcal{A}$ is called a **category of models for an algebraic theory** if there is a finitary monadic functor $U \colon \mathcal{A} \to \mathsf{Set}$.

All of the aforementioned categories are categories of models for an algebraic theory, as their forgetful functors are monadic.

6.6 Limits and colimits in categories of algebras

6.6.1 Definition

A functor is **conservative** if it reflects isomorphisms: Uf is an isomorphism iff f is.

6.6.2 Lemma

Monadic functors are conservative.

Proof

Monadic functors are equivalent to the forgetful functor $U^T\colon \mathcal{C}^T \to \mathcal{C}$ so it is sufficient to check this for U^T. If $f\colon (A,\alpha) \to (B,\beta)$'s underlying map f has an inverse f^{-1}, then that map is an algebra morphism. Indeed, if $\beta T f = f\alpha$ then $\alpha T f^{-1} = f^{-1}\beta$. So if $U^T f$ is an isomorphism, so must f be. ■

6.6.3 Example

It can be shown that the forgetful functor from the category of compact Hausdorff spaces to Set is monadic. Thus a bijective continuous function between compact Hausdorff spaces is a homeomorphism.

6.6.4 Example

Any bijective homomorphism in a category of models for an algebraic theory is an isomorphism. For example, the inverse of a bijective group/monoid/ring homomorphism is also a homomorphism.

6.6.5 Theorem

A monadic functor $U\colon \mathcal{A} \to \mathcal{C}$ creates

(1) any limits $\mathcal{C}$ has, and

(2) any colimits $\mathcal{C}$ has and the monad and its square preserve.

Proof

It is sufficient to prove this for $U^T\colon \mathcal{C}^T \to \mathcal{C}$. For the first point, consider a diagram $D\colon J \to \mathcal{C}^T$; denote the image of $j \in J$ by $(Dj, \gamma_j) \in \mathcal{C}^T$. Now suppose $U^T D\colon J \to \mathcal{C}$ has a limit cone $(\mu_j\colon L \to Dj)_{j\in J}$ in $\mathcal{C}$. We need to lift this limit cone to one of D.

Since (Dj, γ_j) are all algebras, γ assembles into a natural transformation $\gamma\colon TD \Rightarrow D$ in $\mathcal{C}^J$. Consider then the composite $TL \xrightarrow{T\mu} TD \xrightarrow{\gamma} D$, this is a cone over D with summit TL. As $\mu\colon L \Rightarrow D$ is the limit cone, this composite must factor uniquely through it: there exists a unique $\lambda\colon TL \to L$ so that $\gamma T\mu = \mu\lambda$. Then (L,λ) is a T-algebra, as can be shown via diagram chase and the universal property of μ. Furthermore this can be shown to be a limit cone, as desired. ■

6.6.6 Corollary

The inclusion of a reflective subcategory creates all limits. In particular a reflective subcategory of a complete category is complete.

6.6.7 Corollary

Any category which is monadic over Set is complete, with limits created by the monadic forgetful functor.

6.6.8 Example

Consider the functor $\omega^{\mathsf{op}} \to \mathsf{Ring}$ mapping $n \mapsto \mathbb{Z}/p^n\mathbb{Z}$. Then define its limit $\mathbb{Z}_p = \lim_n \mathbb{Z}/p^n\mathbb{Z}$, this is the **$p$-adic integers**. Since $U\colon \mathsf{Ring} \to \mathsf{Set}$ preserves limits,

$$\mathbb{Z}_p = \{(a_n \in \mathbb{Z}/p^n\mathbb{Z})_n \mid a_n \equiv a_m \pmod{p^n}, \text{ for } n \geq m\}$$

as a set. The set functor $\omega^{\mathsf{op}} \to \mathsf{Set}$ also must then preserve the limit cone, which means that the ring structure on $\mathbb{Z}_p$ is one such that the projections $\mathbb{Z}_p \to \mathbb{Z}/p^n\mathbb{Z}$ are all ring morphisms. Thus addition and multiplication are done pointwise.

Note that since the contravariant powerset functor $\mathcal{P}\colon \mathsf{Set}^{\mathsf{op}} \to \mathsf{Set}$ can be shown to be monadic, this tells us that $\mathsf{Set}^{\mathsf{op}}$ is complete and thus Set is cocomplete (as we already know).

Recall that there is a canonical map $\kappa\colon \operatorname{colim} FK \to F \operatorname{colim} K$ for any diagram $K\colon J \to \mathcal{C}$ and $F\colon \mathcal{C} \to \mathcal{D}$ (this is the assembly map).

6.6.9 Proposition

Suppose $\mathcal{C}$ is cocomplete and $U\colon \mathcal{A} \to \mathcal{C}$ is monadic. Then $\mathcal{A}$ is cocomplete iff it has coequalizers.

Proof

All that remains is to show that if $U\colon \mathcal{A} \to \mathcal{C}$ is monadic and has coequalizers then $\mathcal{A}$ has coproducts. We replace U, A with the equivalent $U^T\colon \mathcal{C}^T \to \mathcal{C}$. Let $(A_i, \alpha_i)_i$ be a family of T-algebras, and consider the coequalizer

$$\begin{tikzcd}
\left(T\coprod_i TA_i, \mu_{\coprod_i TA_i}\right) \arrow[rr, "T\coprod_i \alpha_i"'] \arrow[dr, "T\kappa"'] & & \left(T\coprod_i A_i, \mu_{\coprod_i A_i}\right) \arrow[r] & (A, \alpha) \\
& \left(T^2\coprod_i A_i, \mu_{T\coprod_i A_i}\right) \arrow[ur, "\mu_{\coprod_i A_i}"] & &
\end{tikzcd}$$